

10

Angular momentum

In this chapter we shall study the properties of angular momentum, which we have introduced already in Chapter 8. The fundamental property of angular momentum is that it is the infinitesimal generator of rotations. All the results that we shall obtain in this chapter will be more or less direct consequences of this property. In Section 10.1 we explicitly construct a basis of eigenvectors common to $\vec{J}^2$ and J_z , which are compatible Hermitian operators. The rotation of a physical state, which we have already introduced in Chapter 3 for the photon polarization and for spin 1/2, will be studied in the general case in Section 10.2. Section 10.3 is devoted to orbital angular momentum, which originates in the spatial motion of particles. In Section 10.4 we extend the classical results on motion in a central force field to quantum mechanics, and in Section 10.5 we discuss applications to particle decay and excited states. Finally, in Section 10.6 we study the addition of angular momenta.

NB Throughout this chapter we use a system of units in which $\hbar = 1$.

10.1 Diagonalization of $\vec{J}^2$ and J_z

In Chapter 8 we established the commutation relations (8.31) and (8.32) between the various components of angular momentum. Here we give them again in a system of units in which $\hbar = 1$ (we recall that angular momentum has the same dimensions as $\hbar$, which is why the notation is simpler in this system of units):

$$\boxed{[J_x, J_y] = iJ_z, \quad [J_y, J_z] = iJ_x, \quad [J_z, J_x] = iJ_y}, \quad (10.1)$$

or

$$\boxed{[J_k, J_l] = i \sum_m \varepsilon_{klm} J_m}. \quad (10.2)$$

Knowledge of only these commutation relations will permit us to diagonalize the angular momentum, that is, to find the eigenvectors and eigenvalues of suitable combinations of

J_x , J_y , and J_z . Since these three operators do not commute with each other, they cannot be diagonalized simultaneously: the three components of $\vec{J}$ are mutually incompatible physical properties. To choose our combinations of J_x , J_y , and J_z , we observe that $\vec{J}^2$ is a scalar operator (cf. (8.33)) and, according to the result of Section 8.2.3, must commute with the three components of $\vec{J}$:

$$[\vec{J}^2, J_k] = 0, \quad (10.3)$$

as can be verified by explicit calculation (Exercise 10.7.1). The usual choice is to simultaneously diagonalize $\vec{J}^2$ and J_z , and this is often referred to as *quantization of the angular momentum in the z direction*. It is also said that Oz is *chosen as the angular momentum quantization axis*. It is convenient to define the operators $J_{\pm} = J_x \pm iJ_y$ and J_0 as

$$J_{\pm} = J_x \pm iJ_y, \quad J_0 = J_z. \quad (10.4)$$

We can immediately verify the commutation relations and the following identities:

$$[J_0, J_{\pm}] = \pm J_{\pm}, \quad (10.5)$$

$$[J_+, J_-] = 2J_0, \quad (10.6)$$

$$\vec{J}^2 = \frac{1}{2}(J_-J_+ + J_+J_-) + J_0^2, \quad (10.7)$$

$$J_+J_- = \vec{J}^2 - J_0(J_0 - 1), \quad (10.8)$$

$$J_-J_+ = \vec{J}^2 - J_0(J_0 + 1). \quad (10.9)$$

These relations will be useful for the diagonalization. Let $|jm\rangle$ be an eigenvector of $\vec{J}^2$ and J_z , where j labels the eigenvalue of $\vec{J}^2$ and m labels those of J_z . Since $\vec{J}^2$ is a positive operator, its eigenvalues are ≥ 0 . We write them in the form $j(j+1)$ with $j \geq 0$; this notation for the eigenvalues of $\vec{J}^2$ will be justified below. The number m is called the *magnetic quantum number*. In summary:

$$\vec{J}^2|jm\rangle = j(j+1)|jm\rangle, \quad (10.10)$$

$$J_0|jm\rangle = m|jm\rangle. \quad (10.11)$$

According to (10.5), the vectors $J_{\pm}|jm\rangle$ are eigenvectors of J_0 with eigenvalue $m \pm 1$:

$$\begin{aligned} J_0[J_{\pm}|jm\rangle] &= (J_{\pm}J_0 \pm J_{\pm})|jm\rangle = J_{\pm}(m \pm 1)|jm\rangle \\ &= (m \pm 1)[J_{\pm}|jm\rangle]. \end{aligned}$$

Similarly, since $[\vec{J}^2, J_{\pm}] = 0$,

$$\vec{J}^2[J_{\pm}|jm\rangle] = j(j+1)[J_{\pm}|jm\rangle].$$

We have just shown that the vectors $J_{\pm}|jm\rangle$ are eigenvectors of $\vec{J}^2$ with eigenvalue $j(j+1)$ and of J_0 with eigenvalue $m \pm 1$. Moreover, assuming that $|jm\rangle$ is normalized, $\langle jm|jm\rangle = 1$, we can calculate the norm of $J_{+}|jm\rangle$ using (10.9):

$$\begin{aligned} ||J_{+}|jm\rangle||^2 &= \langle jm|J_{-}J_{+}|jm\rangle = \langle jm|\vec{J}^2 - J_0(J_0 + 1)|jm\rangle \\ &= j(j+1) - m(m+1) = (j-m)(j+m+1) \geq 0, \end{aligned} \quad (10.12)$$

and that of $J_{-}|jm\rangle$ using (10.8):

$$\begin{aligned} ||J_{-}|jm\rangle||^2 &= \langle jm|J_{+}J_{-}|jm\rangle = \langle jm|\vec{J}^2 - J_0(J_0 - 1)|jm\rangle \\ &= j(j+1) - m(m-1) = (j+m)(j-m+1) \geq 0. \end{aligned} \quad (10.13)$$

The simultaneous positivity of the two norms is guaranteed only if $-j \leq m \leq j$. Starting from $|jm\rangle$, by repeated application of J_{+} we obtain a series of eigenvectors common to $\vec{J}^2$ and J_0 , labeled by $(j, m+1)$, $(j, m+2)$, etc. These eigenvectors have positive norm as long as $m \leq j$, but the norm becomes negative for $m > j$. The series must therefore terminate, which is possible only if one of the vectors $(J_{+})^n|jm\rangle$ vanishes for an integer value of $n = n_1 + 1$ such that $m + n_1 = j$:

$$J_{+}[(J_{+})^{n_1}|jm\rangle] = 0.$$

The same argument for J_{-} shows that there must exist an integer n_2 such that

$$J_{-}[(J_{-})^{n_2}|jm\rangle] = 0.$$

From the relations

$$j = m + n_1, \quad -j = m - n_2$$

we find that $2j$, and therefore $(2j+1)$, must be an integer, which leads to the diagonalization theorem for $\vec{J}^2$ and J_z .

Theorem. The possible values of j are integers or half-integers: $j = 0, 1/2, 1, 3/2, \dots$. If $|jm\rangle$ is an eigenvector common to $\vec{J}^2$ and J_0 , m necessarily takes one of $(2j+1)$ values:

$$m = -j, -j+1, -j+2, \dots, j-2, j-1, j.$$

When j takes the values $0, 1, 2, \dots$ we have so-called integer angular momentum, and when $j = 1/2, 3/2, \dots$ we have half-integer angular momentum.¹ Let us study the normalization and phase of the vectors $|jm\rangle$. Starting from a vector $|jm\rangle$, by repeated application of J_{+} and J_{-} we construct a series of $(2j+1)$ orthogonal vectors which span a vector

¹ Although half of an even integer is also a half-integer...

subspace of $(2j+1)$ dimensions $\mathcal{E}(j)$ of $\mathcal{H}$. These vectors do not have unit norm, but if we define $|j, m-1\rangle$ by

$$|j, m-1\rangle = [j(j+1) - m(m-1)]^{-1/2} J_- |jm\rangle, \quad (10.14)$$

then $|j, m-1\rangle$ has unit norm according to (10.13). Moreover, using (10.8),

$$\begin{aligned} J_+ J_- |jm\rangle &= [j(j+1) - m(m-1)]^{1/2} J_+ |j, m-1\rangle \\ &= [j(j+1) - m(m-1)] |jm\rangle \end{aligned}$$

or

$$J_+ |j, m-1\rangle = [j(j+1) - m(m-1)]^{1/2} |jm\rangle,$$

and with the replacement $m \rightarrow m+1$ we have

$$J_- |jm\rangle = [j(j+1) - m(m+1)]^{1/2} |j, m+1\rangle. \quad (10.15)$$

The relations (10.14) or (10.15) completely fix the relative phase of the vectors $|j, j\rangle, |j, j-1\rangle, \dots, |j, -j\rangle$. A basis of $\mathcal{E}(j)$ formed from vectors $|jm\rangle$ satisfying (10.14) or (10.15) is called the *standard basis* $|jm\rangle$.

It can happen that knowing (j, m) is not sufficient for uniquely specifying a vector of $\mathcal{H}$: $\vec{J}^2$ and J_z do not form a complete set of compatible physical properties. We shall see an example of this in Section 10.4.2 where we discuss the hydrogen atom. There the values of the (orbital) angular momentum, denoted l , are not sufficient for specifying a bound state; an additional quantum number $n = l+1, l+2, \dots$, called the principal quantum number, must also be given. In general, it is necessary to use a quantum number or a set of supplementary quantum numbers τ to label the eigenvectors $|j, m=j\rangle$ of $\vec{J}^2$ and J_z , and these are normalized by the condition

$$\langle \tau, j, j | \tau', j, j \rangle = \delta_{\tau, \tau'}.$$

By repeated application of J_- we form the standard basis of $\mathcal{E}(\tau, j)$:

$$|\tau, j, j\rangle, |\tau, j, j-1\rangle, \dots, |\tau, j, -j+1\rangle, |\tau, j, -j\rangle.$$

Let us summarize the essential properties of a standard basis $|\tau, jm\rangle$:

$$\vec{J}^2 |\tau, jm\rangle = j(j+1) |\tau, jm\rangle, \quad J_z |\tau, jm\rangle = m |\tau, jm\rangle, \quad (10.16)$$

$$J_+ |\tau, jm\rangle = [j(j+1) - m(m+1)]^{1/2} |\tau, j, m+1\rangle, \quad (10.17)$$

$$J_- |\tau, jm\rangle = [j(j+1) - m(m-1)]^{1/2} |\tau, j, m-1\rangle, \quad (10.18)$$

$$J_+ |\tau, j, j\rangle = 0, \quad J_- |\tau, j, -j\rangle = 0, \quad (10.19)$$

$$\langle \tau', j' m' | \tau, jm \rangle = \delta_{\tau' \tau} \delta_{j' j} \delta_{m' m}. \quad (10.20)$$

In what follows we shall suppress the index τ , as it plays no role in this chapter (except in Section 10.4). The matrix elements of $\vec{J}^2$, J_0 , and $J_{\pm}$ in a standard basis are

$$\langle j' m' | \vec{J}^2 | j m \rangle = j(j+1) \delta_{j'j} \delta_{m'm}, \quad (10.21)$$

$$\langle j' m' | J_0 | j m \rangle = m \delta_{j'j} \delta_{m'm}, \quad (10.22)$$

$$\langle j' m' | J_{\pm} | j m \rangle = [j(j+1) - mm']^{1/2} \delta_{j'j} \delta_{m', m \pm 1}. \quad (10.23)$$

In the subspace $\mathcal{E}(j)$ in which $\vec{J}^2$ has fixed eigenvalue $j(j+1)$, the operators J_0 and $J_{\pm}$ are represented by $(2j+1) \times (2j+1)$ matrices, and the matrix representing J_0 is diagonal. It is instructive (Exercise 10.7.4) to write out these matrices explicitly in the case $j = 1/2$ and recover the 2×2 matrices of spin 1/2 (3.47) as well as those of the case $j = 1$. In the latter case we recover the infinitesimal generators of rotations in three-dimensional space: the transformation law of a vector in $\mathbb{R}^3$ is that of angular momentum $j = 1$. Equation (10.23) gives the following for the infinitesimal generators (Exercise 10.7.4):

$$\boxed{J_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad J_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}}. \quad (10.24)$$

These infinitesimal generators superficially differ in form from the generators T_i found in (8.26). In fact, the two sets are related by the unitary transformation (10.64) which transforms the Cartesian components of $\hat{r}$ into spherical components; see Exercise 10.7.4.

10.2 Rotation matrices

In Chapter 3 we saw how to rotate a spin 1/2. Starting from a state $|+\rangle$ obtained by means of a Stern–Gerlach apparatus in which the magnetic field is parallel to Oz , we know from (3.57) how to construct the state $|+, \hat{n}\rangle$ obtained using a Stern–Gerlach apparatus with magnetic field parallel to $\hat{n}$. We apply to the state $|+\rangle$ a rotation operator $U[\mathcal{R}]$ which transforms $|+\rangle$ into $|+, \hat{n}\rangle$:

$$|+, \hat{n}\rangle = U[\mathcal{R}]|+\rangle = |+\rangle_{\mathcal{R}}.$$

The rotation $\mathcal{R}$ aligns Oz in the direction $\hat{n}$. This rotation is not unique, and we shall see that this nonuniqueness corresponds to an arbitrary phase in the definition of $|+, \hat{n}\rangle$. Another example of the rotation of a physical state was given in Chapter 3 in the case of photon polarization. Starting from a linear polarization state $|x\rangle$, we obtain a linear polarization state $|\theta\rangle$ by applying to the former a rotation operator $U[\mathcal{R}_z(\theta)]$ corresponding to rotation by an angle θ about the photon's direction of propagation Oz (3.29):

$$|\theta\rangle = \exp(-i\theta\Sigma_z)|x\rangle = U[\mathcal{R}_z(\theta)]|x\rangle.$$

In the general case, the state $|\varphi\rangle_{\mathcal{R}}$ transformed by a rotation $\mathcal{R}$ from a state $|\varphi\rangle$ is

$$|\varphi\rangle_{\mathcal{R}} = U[\mathcal{R}]|\varphi\rangle.$$

We now give the explicit matrix form of the rotation operator $U[\mathcal{R}]$ in the basis $|jm\rangle$. The rotation operator $U[\mathcal{R}]$ is expressed as a function of the infinitesimal generators J_x , J_y , and J_z ; cf. (8.30). Since the components of $\vec{J}$ commute with $\vec{J}^2$, the commutator $[U(\mathcal{R}), \vec{J}^2] = 0$ and the matrix elements of U are zero if $j \neq j'$:

$$\langle j' m' | U[\mathcal{R}] | jm \rangle \propto \delta_{jj'}.$$

In the subspace $\mathcal{E}(j)$, the operator $U(\mathcal{R})$ will be represented by a $(2j+1) \times (2j+1)$ matrix denoted $D^{(j)}[\mathcal{R}]$. Its elements are

$$D_{m'm}^{(j)}[\mathcal{R}] = \langle jm' | U[\mathcal{R}] | jm \rangle. \quad (10.25)$$

The matrices $D^{(j)}$ are called *rotation matrices*, or *Wigner matrices*. Let us examine the rotational transformation of a state $|jm\rangle$ giving the vector $|jm\rangle_{\mathcal{R}}$:

$$|jm\rangle_{\mathcal{R}} = U[\mathcal{R}]|jm\rangle = \sum_{m'} |jm'\rangle \langle jm' | U[\mathcal{R}] | jm \rangle,$$

where we have used the fact that in the completeness relation

$$\sum_{j', m'} |j' m'\rangle \langle j' m'| = I$$

only the terms with $j = j'$ contribute. We can then write

$$|jm\rangle_{\mathcal{R}} = \sum_{m'} D_{m'm}^{(j)}[\mathcal{R}] |jm'\rangle. \quad (10.26)$$

Let us recall the group properties of the operators $U[\mathcal{R}]$. In the case of a vector representation (8.12)

$$U[\mathcal{R}_2]U[\mathcal{R}_1] = U[\mathcal{R}_2\mathcal{R}_1], \quad (10.27)$$

while for a spinor representation (8.13)

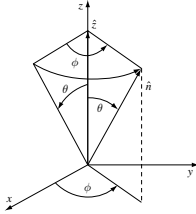
$$U[\mathcal{R}_2]U[\mathcal{R}_1] = \pm U[\mathcal{R}_2\mathcal{R}_1]. \quad (10.28)$$

At the end of this section we shall show that (10.27) corresponds to the case of integer angular momentum and (10.28) to the half-integer case. The multiplication law for rotation matrices is determined by the group property for the operators U :

$$D_{m'm}^{(j)}[\mathcal{R}_2\mathcal{R}_1] = \pm \sum_{m''} D_{m''m}^{(j)}[\mathcal{R}_2] D_{m'm''}^{(j)}[\mathcal{R}_1].$$

Let us return to the study of the rotation which takes Oz to the direction $\hat{n}$ described by the polar and azimuthal angles (θ, ϕ) :

$$\hat{n}_x = \sin \theta \cos \phi, \quad \hat{n}_y = \sin \theta \sin \phi, \quad \hat{n}_z = \cos \theta. \quad (10.29)$$

Fig. 10.1. The rotation $\mathcal{R}(\theta, \phi)$ aligns the axis Oz with $\hat{n}$.

We shall adopt the following *convention* for the rotation: $\mathcal{R}$, denoted $\mathcal{R}(\theta, \phi)$, will be the product of a rotation by an angle θ about Oy followed by one by an angle ϕ about Oz (Fig. 10.1):

$$\boxed{\mathcal{R}(\theta, \phi) = \mathcal{R}_z(\phi)\mathcal{R}_y(\theta)} . \quad (10.30)$$

Using (10.30) and the group law, the rotation operator $U[\mathcal{R}(\theta, \phi)]$ is given as a function of the infinitesimal generators J_y and J_z by

$$U[\mathcal{R}(\theta, \phi)] = e^{-i\phi J_z} e^{-i\theta J_y}, \quad (10.31)$$

and its matrix elements in the basis $|jm\rangle$ are

$$\boxed{D_{m'm}^{(j)}[\mathcal{R}(\theta, \phi)] = \langle jm' | e^{-i\phi J_z} e^{-i\theta J_y} | jm \rangle} . \quad (10.32)$$

This equation can be simplified:

$$D_{m'm}^{(j)}[\mathcal{R}(\theta, \phi)] \equiv D_{m'm}^{(j)}(\theta, \phi) = e^{-im'\phi} \langle jm' | e^{-i\theta J_y} | jm \rangle \quad (10.33)$$

$$= e^{-im'\phi} d_{m'm}^{(j)}(\theta). \quad (10.34)$$

We have defined the matrix $d^{(j)}(\theta)$ as

$$\boxed{d_{m'm}^{(j)}(\theta) = \langle jm' | e^{-i\theta J_y} | jm \rangle} . \quad (10.35)$$

The matrices $d^{(j)}$ satisfy a group property derived from that of the matrices $D^{(j)}$:

$$d_{m'm}^{(j)}(\theta_2 + \theta_1) = \sum_{m''} d_{m''m}^{(j)}(\theta_2) d_{m'm}^{(j)}(\theta_1).$$

There is no sign $\pm$ in this equation because the rotation angle can be greater than 2π .

We have already mentioned the arbitrariness in the choice of rotation (θ, ϕ) ; we could have first rotated by an angle ψ about Oz without changing the final axis $\hat{n}$. In that case the new rotation operator would be

$$U[\mathcal{R}'] = U[\mathcal{R}(\theta, \phi)]e^{-i\psi J_z},$$

and the result (10.26) would acquire the phase factor $\exp(-im\psi)$. The most general definition of the rotation matrices involves three angles, called the Euler angles (ϕ, θ, ψ) , and our convention corresponds to the choice $(\phi, \theta, 0)$.²

In the basis $|jm\rangle$, iJ_y is represented by a real matrix, because according to (10.23) the matrix elements of J_+ and J_- are real and

$$J_y = -\frac{i}{2}(J_+ - J_-).$$

The matrix $\exp(-i\theta J_y)$ is also a real matrix and the group property

$$U^\dagger[\mathcal{R}] = U^{-1}[\mathcal{R}] = U[\mathcal{R}^{-1}]$$

becomes

$$[d^{(j)}(\theta)]^\dagger = [d^{(j)}(-\theta)],$$

which gives the following for the matrix elements:

$$\boxed{d_{m'm}^{(j)}(\theta) = d_{mm'}^{(j)}(-\theta)}. \quad (10.36)$$

There exists another symmetry property (Exercise 10.4.12):

$$\boxed{d_{m'm}^{(j)}(\theta) = (-1)^{m-m'} d_{-m', -m}^{(j)}(\theta)}. \quad (10.37)$$

Finally, it can be shown that the matrices $D^{(j)}$ form a so-called *irreducible* representation of the rotation group, that is, any vector of $\mathcal{E}(j)$ can be obtained from an arbitrary vector of this space by application of a rotation matrix $D^{(j)}$, and any matrix that commutes with all the matrices $D^{(j)}$ is a multiple of the identity matrix.

Whether or not the factor $\pm$ occurs in (10.28) can be checked by studying rotations by 2π , as this factor arises when a rotation by 2π is represented by the operator $-I$ in the space $\mathcal{E}(j)$. Let us consider a rotation by 2π about the z axis:

$$\begin{aligned} \langle jm' | U[\mathcal{R}_z(2\pi)] | jm \rangle &= e^{-2i\pi m} \delta_{m'm} = \delta_{m'm}, & \text{integer } j \\ &= e^{-2i\pi m} \delta_{m'm} = -\delta_{m'm}, & \text{half-integer } j. \end{aligned}$$

² The usual notation for the rotation matrices is

$$D^{(j)}(\theta, \phi) \rightarrow D^{(j)}(\phi, \theta, \psi = 0).$$

Since the choice of axis Oz is arbitrary, the operator rotating by 2π will be I for integer j and $-I$ for half-integer j . However, operators that rotate by 4π are all equal to I for any value of j . Let us examine two successive rotations by angles θ_1 and θ_2 about an axis $\hat{n}$, with

$$\theta_1 + \theta_2 = \theta + 2\pi n, \quad 0 \leq \theta < 2\pi, \quad \text{integer } n \geq 0.$$

From the equations

$$\begin{aligned} e^{-i(\theta_1 + \theta_2)\vec{J} \cdot \hat{n}} &= e^{-i\theta(\vec{J} \cdot \hat{n})} e^{-2i\pi n(\vec{J} \cdot \hat{n})} = e^{-i\theta(\vec{J} \cdot \hat{n})}, \quad \text{integer } j \\ &= (-1)^n e^{-i\theta(\vec{J} \cdot \hat{n})}, \quad \text{half-integer } j, \end{aligned}$$

we find that (10.27) is valid for integer j and (10.28) for half-integer j . In other words, to any rotation $\mathcal{R}$ there correspond two rotation operators of opposite sign for half-integer j and only one for integer j .

Let us check that in the case of spin $1/2$ we recover the matrix $D^{(1/2)}(\theta, \phi)$ already calculated in Chapter 3. The matrix $d^{(1/2)}(\theta)$ according to Exercise 3.3.6 is

$$d^{(1/2)}(\theta) = \exp(-i\theta\sigma_y/2) = \cos \frac{\theta}{2} I - i\sigma_y \sin \frac{\theta}{2},$$

or in explicit form

$$d^{(1/2)}(\theta) = \begin{pmatrix} \cos \theta/2 & -\sin \theta/2 \\ \sin \theta/2 & \cos \theta/2 \end{pmatrix}, \quad (10.38)$$

where the rows and columns are arranged in the order $m = 1/2, -1/2$. Then (10.33) gives the following for the matrix $D^{(1/2)}(\theta, \phi)$:

$$D^{(1/2)}(\theta, \phi) = \begin{pmatrix} e^{-i\phi/2} \cos \theta/2 & -e^{-i\phi/2} \sin \theta/2 \\ e^{i\phi/2} \sin \theta/2 & e^{i\phi/2} \cos \theta/2 \end{pmatrix},$$

in agreement with (3.58).

The rotation matrix $d^{(1)}(\theta)$ for angular momentum $j = 1$ is obtained from the infinitesimal generators (10.24), with the rows and columns arranged in the order $m = 1, 0, -1$ (Exercise 10.7.4):

$$d^{(1)}(\theta) = \begin{pmatrix} \frac{1}{2}(1 + \cos \theta) & -\frac{1}{\sqrt{2}} \sin \theta & \frac{1}{2}(1 - \cos \theta) \\ \frac{1}{\sqrt{2}} \sin \theta & \cos \theta & -\frac{1}{\sqrt{2}} \sin \theta \\ \frac{1}{2}(1 - \cos \theta) & \frac{1}{\sqrt{2}} \sin \theta & \frac{1}{2}(1 + \cos \theta) \end{pmatrix}. \quad (10.39)$$

The reader should verify that the matrices $d^{(1/2)}$ and $d^{(1)}$ possess the symmetry properties (10.36) and (10.37).

10.3 Orbital angular momentum

10.3.1 The orbital angular momentum operator

Let us consider a classical scalar field $\psi(\vec{r})$ and subject it to a rotation $\mathcal{R}_z(\phi)$ by an angle ϕ about Oz , with $\vec{r}' = \mathcal{R}\vec{r}$ being the vector transformed from $\vec{r}$ by this rotation:

$$x' = x \cos \phi - y \sin \phi,$$

$$y' = x \sin \phi + y \cos \phi,$$

$$z' = z.$$

The value of the transformed scalar field $\psi'(\vec{r})$ at the point $\vec{r}'$ must be identical to that of the initial field at the point $\vec{r}$:

$$\psi'(\vec{r}') = \psi(\vec{r}),$$

or

$$\boxed{\psi'(\vec{r}) = \psi(\mathcal{R}^{-1}\vec{r})}. \quad (10.40)$$

This transformation law is correct for a (scalar) classical field, but if $\psi(\vec{r})$ is the wave function of a particle $\psi(\mathcal{R}^{-1}\vec{r})$ and $\psi'(\vec{r})$ can a priori differ by a phase:

$$\psi'(\vec{r}) = e^{i\theta(\vec{r})} \psi(\mathcal{R}^{-1}\vec{r})$$

(see the discussion following (9.17)). We know only that $|\psi'(\vec{r}')| = |\psi(\vec{r})|$, and our goal is to show that the phase factor that might arise is actually absent. The vector $U(\mathcal{R})|\vec{r}\rangle$ physically represents an eigenstate of the position operator $\vec{R}$, obtained from the eigenstate $|\vec{r}\rangle$ of $\vec{R}$ by a rotation $U(\mathcal{R})$. Let us show this explicitly using the fact that $\vec{R}$ is a vector operator whose components X_k transform as the components of V in (8.34):

$$\begin{aligned} X_k[U(\mathcal{R})|\vec{r}\rangle] &= U(\mathcal{R})U^{-1}(\mathcal{R})X_kU(\mathcal{R})|\vec{r}\rangle \\ &= U(\mathcal{R})\left(\sum_l \mathcal{R}_{kl}X_l\right)|\vec{r}\rangle = U(\mathcal{R})\left(\sum_l \mathcal{R}_{kl}x_l\right)|\vec{r}\rangle \\ &= (\mathcal{R}\vec{r})_k[U(\mathcal{R})|\vec{r}\rangle], \end{aligned}$$

which shows that the state vector $|\mathcal{R}\vec{r}\rangle$ can be defined, that is, its phase can be fixed, as

$$|\mathcal{R}\vec{r}\rangle \equiv U(\mathcal{R})|\vec{r}\rangle. \quad (10.41)$$

If $|\psi'\rangle$ is the transform of $|\psi\rangle$ by $U(\mathcal{R})$, $|\psi'\rangle = U(\mathcal{R})|\psi\rangle$, then

$$\begin{aligned} \psi'(\vec{r}) &= \langle \vec{r}|\psi'\rangle = \langle \vec{r}|U(\mathcal{R})|\psi\rangle = \langle U^\dagger(\mathcal{R})\vec{r}|\psi\rangle \\ &= \langle U^{-1}(\mathcal{R})\vec{r}|\psi\rangle = \langle \mathcal{R}^{-1}\vec{r}|\psi\rangle = \psi(\mathcal{R}^{-1}\vec{r}), \end{aligned}$$

which demonstrates (10.40). At first sight the argument $\mathcal{R}^{-1}$ in (10.40), which can also be written as

$$[U(\mathcal{R})\psi](\vec{r}) = \psi(\mathcal{R}^{-1}\vec{r}),$$

may seem surprising, but we have already encountered a similar situation in the case of translations in (9.15), which in three dimensions with $\hbar = 1$ is written as

$$\left[e^{-i\vec{p}\cdot\vec{a}} \psi \right](\vec{r}) = \psi(\vec{r} - \vec{a}),$$

even though³

$$e^{-i\vec{p}\cdot\vec{a}}|\vec{r}\rangle = |\vec{r} + \vec{a}\rangle.$$

The function $\psi(\vec{r})$ transformed by a translation $\vec{a}$ is $\psi(\vec{r} - \vec{a})$ and not $\psi(\vec{r} + \vec{a})$! If the rotation angle ϕ becomes infinitesimal for a rotation about Oz , then

$$U[\mathcal{R}_z(\phi)] \simeq I - i\phi J_z,$$

and according to (10.40)

$$\begin{aligned} [(I - i\phi J_z)\psi](\vec{r}) &\simeq \psi(x + y\phi, -x\phi + y, z) \\ &\simeq \psi(\vec{r}) + \phi \left(y \frac{\partial \psi}{\partial x} - x \frac{\partial \psi}{\partial y} \right) \\ &= \psi(\vec{r}) - i\phi (XP_y - YP_x)\psi, \end{aligned}$$

from which we find

$$[J_z\psi](\vec{r}) = [(XP_y - YP_x)\psi](\vec{r}) = [(\vec{R} \times \vec{P})_z\psi](\vec{r}) = (\vec{L}_z\psi)(\vec{r}). \quad (10.42)$$

The angular momentum operator of the particle described by a wave function $\psi(\vec{r})$ is called the *orbital angular momentum* (because it is associated with the motion of the particle in a spatial orbit), and is in general denoted $\vec{L}$:

$$\vec{L} = \vec{R} \times \vec{P}. \quad (10.43)$$

The operator $\vec{L}$ has been constructed as the infinitesimal generator of rotations and necessarily satisfies the angular momentum commutation relations (10.1) or (10.2):

$$[L_j, L_k] = i \sum_l \varepsilon_{jkl} L_l. \quad (10.44)$$

These relations can be verified by explicit calculation using the canonical commutation relations (8.45); see Exercise 10.7.5. We use $|lm\rangle$ to denote the eigenvectors of $\vec{L}^2$ and L_z :

$$\vec{L}^2|lm\rangle = l(l+1)|lm\rangle, \quad (10.45)$$

$$L_z|lm\rangle = m|lm\rangle. \quad (10.46)$$

These equations can be transformed into differential equations by writing the operators L_j as differential operators acting in $L^{(2)}(\mathbb{R}^3)$. The calculation is lengthy if we make the change of variables $(x, y, z) \rightarrow (r, \theta, \phi)$, but it is simplified if we use the fact that the L_i

³ We note that this equation fixes the phase of the vector $|\vec{r} + \vec{a}\rangle$ relative to that of $|\vec{r}\rangle$, in the same way as (10.41) fixes the phase of $|\mathcal{R}\vec{r}\rangle$ relative to that of $|\vec{r}\rangle$.

are infinitesimal generators of rotations. The case of L_z is particularly simple. Considering ψ as a function of (r, θ, ϕ) , we have

$$(e^{-i\alpha L_z} \psi)(r, \theta, \phi) = \psi(r, \theta, \phi - \alpha),$$

and taking α to be infinitesimal,

$$([I - i\alpha L_z] \psi)(r, \theta, \phi) = \psi(r, \theta, \phi) - \alpha \frac{\partial \psi}{\partial \phi}$$

or $L_z \psi = -i(\partial \psi / \partial \phi)$. The calculation of L_x and L_y takes a few more lines, because both θ and ϕ vary in a rotation about Ox or Oy . The result is (Exercise 10.7.5)

$$L_z = -i \frac{\partial}{\partial \phi}, \quad (10.47)$$

$$L_{\pm} = ie^{\pm i\phi} \left(\cot \theta \frac{\partial}{\partial \phi} \mp i \frac{\partial}{\partial \theta} \right), \quad (10.48)$$

$$\vec{L}^2 = - \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left[\sin \theta \frac{\partial}{\partial \theta} \right] + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right). \quad (10.49)$$

The operators L_j depend only on angles and not on r , hence the name *angular momentum*. The eigenfunctions of $\vec{L}^2$ and L_z depend only on the angles θ and ϕ or, equivalently, on $\hat{r}$. These eigenfunctions are called the *spherical harmonics*:

$$Y_l^m(\theta, \phi) = Y_l^m(\hat{r}) = \langle \hat{r} | lm \rangle. \quad (10.50)$$

Equations (10.45) and (10.46) become

$$[\vec{L}^2 Y_l^m](\hat{r}) = \langle \hat{r} | \vec{L}^2 | lm \rangle = l(l+1) Y_l^m(\hat{r}), \quad (10.51)$$

$$[L_z Y_l^m](\hat{r}) = \langle \hat{r} | L_z | lm \rangle = m Y_l^m(\hat{r}), \quad (10.52)$$

while (10.15) is written as

$$[L_{\pm} Y_l^m](\hat{r}) = \langle \hat{r} | L_{\pm} | lm \rangle = [l(l+1) - m(m \pm 1)]^{1/2} Y_l^{m \pm 1}(\hat{r}).$$

Equation (10.52) becomes, using (10.47),

$$[L_z Y_l^m](\theta, \phi) = -i \frac{\partial}{\partial \phi} Y_l^m(\theta, \phi) = m Y_l^m(\theta, \phi),$$

which implies that

$$Y_l^m(\theta, \phi) = e^{im\phi} f_l^m(\theta). \quad (10.53)$$

The transformation law (10.40) shows that in a rotation by 2π the wave function is unchanged, and so no minus sign is introduced. This implies that *orbital angular momenta are always integers*.

A simple and important application is the spherical rotator. We consider a diatomic molecule rotating about its center of mass, taken to be the coordinate origin (Fig. 10.2 and Exercise 1.6.1). Its moment of inertia is $I = \mu r_0^2$, where μ is the reduced mass and

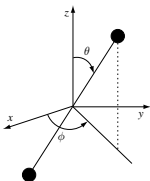


Fig. 10.2. The spherical rotator.

r_0 is the distance between the nuclei (the electron contribution is negligible). If ω is the angular velocity of the rotation, the classical Hamiltonian H_{cl} is

$$H_{cl} = \frac{1}{2} I \omega^2 = \frac{1}{2} \frac{(I\omega)^2}{I} = \frac{l^2}{2I},$$

where $l = I\omega$ is the angular momentum. The quantum version of the Hamiltonian is

$$H = \frac{\vec{L}^2}{2I},$$

and the energies are

$$E_l = \frac{l(l+1)}{2I}. \quad (10.54)$$

The eigenfunctions are the $Y_l^m(\theta, \phi)$, where the angles θ and ϕ specify the orientation of the line joining the two nuclei; $Y_l^m(\theta, \phi)$ is the amplitude for finding this line oriented in the direction (θ, ϕ) . The spectrum of rotational levels is given in Fig. 10.3, and well reproduces the experimental results for the spectra of diatomic molecules.

10.3.2 Properties of the spherical harmonics

Let us now summarize, in some cases without proof, the properties of the spherical harmonics that are most frequently used.

1. Basis on the unit sphere

The spherical harmonics form an orthonormal basis for square-integrable functions on the unit sphere $\vec{r}^2 = 1$:

$$\int \sin \theta d\theta d\phi [Y_l^{m'}(\theta, \phi)]^* Y_l^m(\theta, \phi) = \int d\Omega [Y_l^{m'}(\theta, \phi)]^* Y_l^m(\theta, \phi) = \delta_{l'l} \delta_{m'm}. \quad (10.55)$$

We shall frequently use the notation $\Omega = (\theta, \phi)$ and

$$d\Omega = \sin \theta d\theta d\phi = d^2 \hat{r}. \quad (10.56)$$

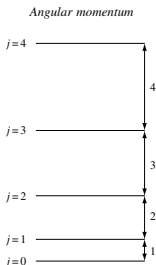


Fig. 10.3. Spectrum of the spherical rotator. The j th level is separated from the $(j-1)$ th level by an amount j/I , or $\hbar^2 j/I$ if $\hbar$ is restored.

If a function $f(\theta, \phi)$ is square-integrable on the unit sphere, we can write down an expansion analogous to a Fourier series:

$$f(\theta, \phi) = \sum_{l,m} c_{lm} Y_l^m(\theta, \phi),$$

$$c_{lm} = \int d\Omega [Y_l^m(\theta, \phi)]^* f(\theta, \phi). \quad (10.57)$$

2. Relation to the Legendre polynomials

One definition of the Legendre polynomials $P_l(u)$ is

$$P_l(u) = \frac{1}{2^l l!} \frac{d^l}{du^l} (u^2 - 1)^l, \quad (10.58)$$

where $P_l(u)$ is a polynomial of degree l and parity $(-1)^l$:

$$P_l(-u) = (-1)^l P_l(u).$$

The Legendre polynomials form a complete set of orthogonal polynomials in the interval $[-1, +1]$. The first few Legendre polynomials are

$$P_0(u) = 1, \quad P_1(u) = u, \quad P_2(u) = \frac{1}{2}(3u^2 - 1). \quad (10.59)$$

The associated Legendre functions $P_l^m(u)$ are defined as

$$P_l^m(u) = (1 - u^2)^{m/2} \frac{d^m}{du^m} P_l(u), \quad P_l^0(u) = P_l(u), \quad (10.60)$$

and it can be shown that the spherical harmonics are related to the P_l^m as

$$Y_l^m(\theta, \phi) = (-1)^m \left[\frac{(2l+1)}{4\pi} \frac{(l-m)!}{(l+m)!} \right]^{1/2} P_l^m(\cos \theta) e^{im\phi}, \quad m > 0, \quad (10.61)$$

$$Y_l^m(\theta, \phi) = (-1)^m [Y_l^{-m}(\theta, \phi)]^*, \quad m < 0.$$

According to (10.53), Y_l^0 is independent of ϕ and proportional to $P_l(\cos \theta)$:

$$Y_l^0(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi}} P_l(\cos \theta). \quad (10.62)$$

As a special case, we write down the Y_l^m for $l = 0$ and $l = 1$:

$$l = 0: Y_0^0 = \sqrt{\frac{1}{4\pi}},$$

$$l = 1: Y_1^0 = \sqrt{\frac{3}{4\pi}} \cos \theta = \sqrt{\frac{3}{4\pi}} \hat{r}_0, \quad (10.63)$$

$$Y_1^\pm = \mp \sqrt{\frac{3}{8\pi}} e^{\pm i\phi} \sin \theta = \sqrt{\frac{3}{4\pi}} \hat{r}_{\pm 1}.$$

Up to the normalization factor $\sqrt{3/4\pi}$ the Y_1^m are just the *spherical components* of the unit vector $\hat{r}$:

$$\hat{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta),$$

$$Y_1^0 = \sqrt{\frac{3}{4\pi}} \hat{r}_0, \quad Y_1^\pm = \mp \sqrt{\frac{3}{4\pi}} \frac{\hat{r}_x \pm i\hat{r}_y}{\sqrt{2}} = \sqrt{\frac{3}{4\pi}} \hat{r}_{\pm 1}. \quad (10.64)$$

These expressions justify the phase conventions used for right- and left-handed polarization in (3.11).

3. Transformation under rotation

Multiplying (10.26) for $j = l$ on the left by the bra $\langle \hat{r} |$, we find

$$Y_l^m(\mathcal{R}^{-1}\hat{r}) = \sum_{m'} D_{m'm}^{(l)}(\mathcal{R}) Y_l^{m'}(\hat{r}). \quad (10.65)$$

We can also obtain (Exercise 10.7.6) a relation between the spherical harmonics and the rotation matrices:

$$D_{m0}^{(l)}(\theta, \phi) = \sqrt{\frac{4\pi}{2l+1}} [Y_l^m(\theta, \phi)]^*. \quad (10.66)$$

From these two equations we can derive the addition theorem for the spherical harmonics. Taking $\hat{r}$ in the direction given by the polar angles (α, β) , let $\mathcal{R}$ be the rotation by angles (θ, ϕ) aligning $\hat{z}$ with $\hat{n}$ and Θ be the angle between $\hat{r}$ and the direction defined by the angles (θ, ϕ) (Fig. 10.4):

$$\cos \Theta = \cos \alpha \cos \theta + \sin \alpha \sin \theta \cos(\beta - \phi).$$

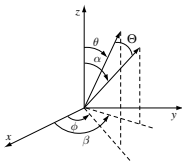


Fig. 10.4. Angular configuration in (10.67).

The angle Θ between $\mathcal{R}^{-1}\hat{r}$ and the z axis is the same as the angle between $\hat{n}$ and $\hat{r}$. It is then sufficient to take $m = 0$ in (10.65) to obtain

$$P_l(\cos \Theta) = \frac{4\pi}{2l+1} \sum_{m=-l}^l [Y_l^m(\theta, \phi)]^* Y_l^m(\alpha, \beta). \quad (10.67)$$

4. Parity of the spherical harmonics

The parity operator Π defined in Section 8.3.3 acts on a wave function $\psi(\vec{r})$ as

$$[\Pi\psi](\vec{r}) = \psi(-\vec{r}). \quad (10.68)$$

Π commutes with the orbital angular momentum $\vec{L}$ and, more generally, with $\vec{J}$. In fact, the representation matrix of the parity operator in three-dimensional space $\mathbb{R}^3$ is the matrix $-I$, which commutes with any rotation matrix $\mathcal{R}$, from which we infer

$$[U(\mathcal{R}), \Pi] = 0 \Rightarrow [\vec{J}, \Pi] = 0 \quad \text{and} \quad [\vec{L}, \Pi] = 0. \quad (10.69)$$

This implies the equations

$$\vec{L}^2 \Pi Y_l^m = \Pi \vec{L}^2 Y_l^m = l(l+1) \Pi Y_l^m,$$

$$L_z \Pi Y_l^m = \Pi L_z Y_l^m = m \Pi Y_l^m,$$

which show that ΠY_l^m is proportional to Y_l^m :

$$\Pi Y_l^m = \alpha(l, m) Y_l^m.$$

Y_l^m is therefore an eigenfunction of Π , and since $\Pi^2 = I$, $\alpha(l, m) = \pm 1$. Let us show that $\alpha(l, m)$ is in fact independent of m using the fact that L_+ commutes with Π :

$$\begin{aligned} L_+ \Pi Y_l^m &= \alpha(l, m) L_+ Y_l^m = \alpha(l, m) [l(l+1) - m(m+1)]^{1/2} Y_l^{m+1} \\ &= \Pi L_+ Y_l^m = [l(l+1) - m(m+1)]^{1/2} \Pi Y_l^{m+1} \\ &= [l(l+1) - m(m+1)]^{1/2} \alpha(l, m+1) Y_l^{m+1}, \end{aligned}$$

which implies that $\alpha(l, m+1) = \alpha(l, m)$. Therefore, $\alpha(l, m)$ is independent of m and

$$[\Pi Y_l^m](\hat{r}) = \alpha(l) Y_l^m(\hat{r}) = Y_l^m(-\hat{r}).$$

The transformation $\hat{r} \rightarrow -\hat{r}$ corresponds to

$$\theta \rightarrow \pi - \theta, \quad \phi \rightarrow \phi + \pi. \quad (10.70)$$

If $m = 0$, then $Y_l^0 \propto P_l(\cos \theta)$; using (10.62) and $P_l(-u) = (-1)^l P_l(u)$, we find $\alpha(l) = (-1)^l$ and

$$Y_l^m(\theta, \phi) = (-1)^l Y_l^m(\pi - \theta, \phi + \pi) \quad \text{or} \quad Y_l^m(\hat{r}) = (-1)^l Y_l^m(-\hat{r}). \quad (10.71)$$

10.4 Particle in a central potential

10.4.1 The radial wave equation

We shall use the preceding results to show that the three-dimensional Schrödinger equation, which is a partial differential equation, can be reduced to an ordinary differential equation when the potential is central, that is, invariant under rotation:

$$V(\vec{r}) = V(|\vec{r}|) = V(r).$$

In this case, since the kinetic energy is a scalar operator, the full Hamiltonian for a particle of mass M

$$H = \frac{\vec{P}^2}{2M} + V(r) \quad (10.72)$$

is invariant under rotation: $[H, \vec{J}] = 0$. Our problem involves only the orbital angular momentum, since the only operators at our disposal are $\vec{P}$ and $\vec{R}$:

$$[H, \vec{L}] = 0 \quad \text{or} \quad [H, L_x] = [H, L_y] = [H, L_z] = 0. \quad (10.73)$$

In the space $L_r^{(2)}(\mathbb{R}^3)$ the kinetic energy operator is proportional to the Laplacian ∇^2 :

$$-\vec{P}^2 = -(-i\nabla)^2 = \nabla^2 = \frac{1}{r} \frac{\partial^2}{\partial r^2} r + \frac{1}{r^2} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right], \quad (10.74)$$

where we have written the Laplacian in polar coordinates. Comparing with (10.49), we recognize in the operator $\vec{L}^2$ the angular part of the Laplacian:

$$\nabla^2 = \frac{1}{r} \frac{\partial^2}{\partial r^2} r - \frac{1}{r^2} \vec{L}^2. \quad (10.75)$$

This equation confirms the commutation relation $[H, \vec{L}] = 0$, since $[\vec{L}^2, \vec{L}] = 0$ and the radial part of the Laplacian, which does not depend on angles, obviously commutes with $\vec{L}$. We can therefore write the Hamiltonian (10.72) as

$$H = -\frac{1}{2M} \frac{1}{r} \frac{\partial^2}{\partial r^2} r + \frac{1}{2Mr^2} \vec{L}^2 + V(r). \quad (10.76)$$

Owing to these commutation relations, we know that it is possible to simultaneously diagonalize H , $\vec{L}^2$, and L_z . Let $\psi_{lm}(\vec{r})$ be an eigenfunction common to these three operators. Since there is only one spherical harmonic (l, m) , if

$$\vec{L}^2 \psi_{lm} = l(l+1) \psi_{lm} \quad \text{and} \quad L_z \psi_{lm} = m \psi_{lm},$$

then ψ_{lm} must be proportional to Y_l^m .⁴

$$\psi_{lm}(r, \theta, \phi) = f_l(r) Y_l^m(\theta, \phi) = \frac{u_l(r)}{r} Y_l^m(\theta, \phi). \quad (10.77)$$

It is convenient to factorize $1/r$; $u_l(r)$ is the *radial wave function*. Let us examine the action of H on ψ_{lm} :

$$H\psi_{lm}(r, \theta, \phi) = \left[-\frac{1}{2M} \frac{1}{r} \frac{\partial^2}{\partial r^2} u_l(r) + \left(\frac{l(l+1)}{2Mr^2} + V(r) \right) \frac{u_l(r)}{r} \right] Y_l^m(\theta, \phi).$$

The eigenvalue equation

$$H\psi_{lm} = E_l \psi_{lm}$$

becomes the radial equation

$$\boxed{\left[-\frac{1}{2M} \frac{d^2}{dr^2} + \frac{l(l+1)}{2Mr^2} + V(r) \right] u_l(r) = E_l u_l(r)}. \quad (10.78)$$

The radial wave function and the energy are labeled by only the index l and not m , because according to (10.78) they are independent of m . Each value of the energy will therefore be at least $(2l+1)$ -fold degenerate. This could have been foreseen from the commutation relation $[H, L_{\pm}] = 0$. If

$$H\psi_{lm} = E_{lm} \psi_{lm},$$

by reasoning similar to that which enabled us to show that $\alpha(l, m)$ is independent of m , we deduce that E_{lm} is also independent of m (Exercise 10.7.7). For each value of the angular momentum l , or for each *partial wave* l , we have reduced the Schrödinger equation to an ordinary differential equation in a single variable r . Following historical tradition, the partial waves are labeled $s, p, d, f, g, h, \dots$:

$$l = 0 : s \text{ wave}, \quad l = 1 : p \text{ wave}, \quad l = 2 : d \text{ wave}, \quad l = 3 : f \text{ wave},$$

and so on in alphabetical order: $l = 4$: g wave, etc. In each partial wave, (10.78) shows that the potential $V(r)$ must be replaced by an *effective potential* $V_l(r)$ (Fig. 10.5):

$$V_l(r) = V(r) + \frac{l(l+1)}{2Mr^2}. \quad (10.79)$$

⁴ We anticipate the fact, proved a few lines later, that f_l is independent of m .

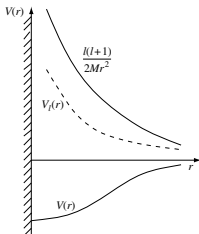


Fig. 10.5. An effective potential. The solid lines represent the potential $V(r)$ and the centrifugal barrier $l(l+1)/2Mr^2$, and the dashed lines represent their sum, the effective potential $V_l(r)$ in the partial wave of angular momentum l .

The term $l(l+1)/2Mr^2$ is called the *centrifugal barrier* term. It is also present in classical mechanics, where the energy can be written as

$$E = \frac{1}{2}M\vec{v}^2 + V(r) = \frac{1}{2}M(v_r^2 + \omega^2 r^2) + V(r),$$

where v_r is the radial velocity and ω the angular velocity. Since⁵ $l = M\omega r^2$ and $\vec{l}$ is constant in the case of a central force, we have

$$E = \frac{1}{2}Mv_r^2 + \frac{l^2}{2Mr^2} + V(r) = \frac{1}{2}Mv_r^2 + V_l(r).$$

The term $l^2/2Mr^2$ corresponds to the centrifugal force:

$$-\frac{d}{dr}\left(\frac{l^2}{2Mr^2}\right) = \frac{l^2}{Mr^3} = M\omega^2 r.$$

This term tends to push the particle away from the force center in the rotating frame and corresponds to a repulsive potential. In quantum mechanics we replace the operator $\vec{L}^2$ by its eigenvalue $l(l+1)$ for each value of l , and to the potential $V(r)$ we add the repulsive potential $l(l+1)/2Mr^2$.

Not all functions $\psi_{lm}(\vec{r})$ of the type (10.77) with $u_l(r)$ a solution of (10.78) are physically acceptable. If the function $\psi_{lm}(\vec{r})$ represents a bound state, it must satisfy the normalization condition

$$\int d^3r |\psi_{lm}(\vec{r})|^2 = 1. \quad (10.80)$$

⁵ Following our usual convention, lower-case letters denote classical quantities (numbers) or quantum numbers.

If $\psi_{lm}(\vec{r})$ represents a scattering state, behavior corresponding to a plane wave plus a spherical wave at infinity $\exp(\pm ikr)/r$ is acceptable [cf. (10.81)]. In the case of a bound state, (10.78) in general possesses several solutions for l fixed. In fact, since $0 \leq r < \infty$ this equation is identical to that of the one-dimensional problem in the range $[0, +\infty]$ with $V_l(r)$ (10.79) as the effective potential. The radial wave function and the energy are labeled by an additional quantum number n' , $n' = 0, 1, 2, \dots$, and denoted as $u_{n'l}(r)$ and $E_{n'l}$. If the potential $V(r)$ is sufficiently smooth, it can be shown that n' is equal to the number of zeros, also called nodes, of the radial wave function $u_{n'l}(r)$ (cf. Section 9.3.3). The quantum number n' classifies the values of the energy in increasing order:

$$n'_1 > n'_2 \Rightarrow E_{n'_1 l} > E_{n'_2 l}.$$

In Chapter 12 we shall see that the wave functions of scattering states are labeled by the wave vector $\vec{k}$:

$$r \rightarrow \infty : \psi_{\vec{k}}(\vec{r}) \simeq e^{i\vec{k} \cdot \vec{r}} + f(\theta, \phi) \frac{e^{\pm ikr}}{r}. \quad (10.81)$$

It is possible to analyze the behavior of the wave functions $u_{n'l}(r)$ for $r \rightarrow 0$. In all cases of physical interest the centrifugal barrier term is the most singular term when $r \rightarrow 0$ and it controls the behavior of $u_{nl}(r)$ in this limit. If we assume a power-law behavior⁶

$$r \rightarrow 0 : u_l(r) \propto r^\alpha$$

and substitute it into (10.78), for the two most singular terms in $r^{\alpha-2}$ we obtain

$$-\frac{1}{2M}\alpha(\alpha-1)r^{\alpha-2} + \frac{l(l+1)}{2M}r^{\alpha-2} = 0,$$

which implies that

$$\alpha(\alpha-1) = l(l+1),$$

i.e., $\alpha = l+1$ or $\alpha = -l$. The second value is excluded because the integral (10.80) diverges at the origin unless $l=0$. However, for $l=0$ a solution $u_0(r) \propto \text{const.}$, or $\psi_l(\vec{r}) \propto 1/r$, although normalizable, is not acceptable because it cannot be a solution of the Schrödinger equation owing to

$$\nabla^2 \frac{1}{r} = -4\pi\delta(\vec{r}).$$

In summary, the behavior of the radial wave functions for $r \rightarrow 0$ is

$$\boxed{r \rightarrow 0 : u_l(r) \propto r^{l+1}}. \quad (10.82)$$

The radial wave function vanishes at the origin. This can be seen intuitively: since $0 \leq r < \infty$, it is as though there were an infinite potential barrier at $r=0$, and we know that in this case (see Section 9.3.2) the wave function must vanish. Nevertheless, the

⁶ The power law giving the behavior at the origin is independent of the quantum numbers n' and l , and so we suppress them.

solutions involving r^{-l} may be useful in solving the Schrödinger equation in a region where r is strictly positive.

The example of the hydrogen atom, which is studied in the following subsection, leads to a redefinition of the radial quantum number, which becomes the principal quantum number:

$$n' \rightarrow n = n' + l + 1. \quad (10.83)$$

10.4.2 The hydrogen atom

The results of the preceding subsection can be used to calculate the energy levels and wave functions of the hydrogen atom, which is one of the few physical problems for which an analytic solution is available. The mass M in (10.78) is the electron mass m_e , or, more precisely, the reduced mass μ (Exercise 8.5.6):

$$\mu = \frac{m_e m_p}{m_e + m_p} \simeq m_e, \quad (10.84)$$

where m_p is the proton mass. However, we shall use m_e rather than μ in the equations in order to emphasize the order of magnitude of the masses which are relevant to this problem. The potential $V(r)$ is the attractive Coulomb potential between the electron and the proton:

$$V(r) = -\frac{q_e^2}{4\pi\epsilon_0 r} = -\frac{e^2}{r}, \quad (10.85)$$

and (10.78) becomes

$$\left[-\frac{1}{2m_e} \frac{d^2}{dr^2} + \frac{l(l+1)}{2m_e r^2} - \frac{e^2}{r} \right] u_{nl}(r) = E_{nl} u_{nl}(r). \quad (10.86)$$

In physics it is always advisable to make equations dimensionless by an appropriate change of variable. In the present problem the natural unit of length is the Bohr radius (1.34) $a_0 = 1/m_e e^2$, and the natural unit of energy is the Rydberg (1.35) $R_\infty = e^2/2a_0 = m_e e^4/2$.⁷ This suggests that we define the dimensionless quantities x and ε_{nl} :

$$x = \frac{r}{a_0} = m_e e^2 r, \quad \varepsilon_{nl} = -\frac{E_{nl}}{R_\infty} = -\frac{2a_0 E_{nl}}{e^2}. \quad (10.87)$$

In what follows we limit ourselves to bound states for which $E_{nl} < 0$ and therefore $\varepsilon_{nl} > 0$, whence the choice of the minus sign. Also defining

$$v_{nl}(x) = u_{nl}(r) = u_{nl}(a_0 x),$$

after simplification by $(2m_e a_0^2)^{-1}$ we obtain

$$\left[-\frac{d^2}{dx^2} + \frac{l(l+1)}{x^2} - \frac{2}{x} \right] v_{nl}(x) = -\varepsilon_{nl} v_{nl}(x). \quad (10.88)$$

⁷ We recall that we have chosen a system of units in which $\hbar = 1$. If $\hbar$ is restored, then $a_0 = \hbar^2/m_e e^2$ and $R_\infty = m_e e^4/2\hbar^2$.

We shall limit ourselves to finding the solution in the case $l = 0$, that is, in the s wave, and leave the general case to Exercise 10.7.9. To simplify the notation, we set

$$v_{n0}(x) = v(x), \quad \varepsilon_{n0} = \varepsilon,$$

and (10.88) becomes

$$\frac{d^2 v(x)}{dx^2} = \left(\varepsilon - \frac{2}{x} \right) v(x).$$

We know from the preceding subsection that $v(x) \propto x$ for $x \rightarrow 0$. Let us now find the dominant behavior for $x \rightarrow \infty$ neglecting the term involving $2/x$. We then have⁸

$$v(x) \sim \exp(\pm \sqrt{\varepsilon} x).$$

The $\exp(\sqrt{\varepsilon} x)$ behavior is unacceptable because the wave function will not be normalizable owing to the exponential divergence. The only possible behavior is $\exp(-\sqrt{\varepsilon} x)$. In order to include the information contained in the behavior at infinity, we define a new function $f(x)$ as

$$v(x) = e^{-\alpha x} f(x), \quad \alpha^2 = \varepsilon.$$

This change of function transforms the differential equation for $v(x)$ into

$$\frac{d^2 f}{dx^2} - 2\alpha \frac{df}{dx} + \frac{2}{x} f = 0. \quad (10.89)$$

Let us seek $f(x)$ in the form of a series in powers of x . Since we know that $f(x) \propto x$ for $x \rightarrow 0$,

$$f(x) = \sum_{k=1}^{\infty} a_k x^k. \quad (10.90)$$

Equation (10.89) determines a recursion relation for the coefficients a_k :

$$\sum_{k=1}^{\infty} k(k-1)a_k x^{k-2} - 2\alpha \sum_{k=1}^{\infty} k a_k x^{k-1} + 2 \sum_{k=1}^{\infty} a_k x^{k-1} = 0.$$

Noting that for $k = 1$ the first term in the preceding equation vanishes and relabeling k , we have

$$\sum_{k=1}^{\infty} [k(k+1)a_{k+1} - 2(\alpha k - 1)a_k] x^{k-1} = 0. \quad (10.91)$$

The cancellation of the coefficient of x^{k-1} gives a relation between a_{k+1} and a_k :

$$a_{k+1} = \frac{2(\alpha k - 1)}{k(k+1)} a_k.$$

⁸ In fact, this behavior is determined only up to a multiplicative polynomial.

If we arbitrarily fix a_1 , all the a_k can be derived from a_1 . For $k \gg 1$ the recursion relation is approximately

$$a_{k+1} \simeq \frac{2\alpha}{k} a_k \Rightarrow a_k \simeq \frac{(2\alpha)^k}{k!} a_1$$

and

$$\sum_{k=1}^{\infty} a_k x^k \sim \sum_{k=1}^{\infty} \frac{(2\alpha)^k}{k!} a_1 x^k \sim a_1 e^{2\alpha x}.$$

This implies that for $x \rightarrow \infty$

$$v(x) \sim e^{2\alpha x} e^{-\alpha x} \sim a_1 e^{\alpha x},$$

which makes the wave function non-normalizable. The only way to avoid the exponential divergence is to have the series (10.90) terminate at some integer $k = n$, which can happen only if $\alpha n = 1$. The possible values of ε then are labeled by an integer n :

$$\varepsilon_n = \alpha^2 = \frac{1}{n^2},$$

as are those of the energy:

$$E_n = E_{n0} = -\frac{me^4}{2} \frac{1}{n^2} = -\frac{R_{\infty}}{n^2}. \quad (10.92)$$

Exercise 10.7.9 shows that the possible energies for $l \neq 0$ have the form

$$E_{nl} = -\frac{R_{\infty}}{n^2}, \quad n = l+1, l+2, \dots \quad (10.93)$$

The first two ($n = 1, 2$) radial wave functions $v_{n0}(x)$ of the bound states of the hydrogen atom in the s wave, normalized to unity, are

$$v_{10}(x) = 2xe^{-x}, \quad (10.94)$$

$$v_{20}(x) = \frac{1}{\sqrt{2}} x \left(1 - \frac{x}{2}\right) e^{-x/2}. \quad (10.95)$$

The radial wave function in the state $n = 2, l = 1$ (the p wave) is

$$v_{21}(x) = \frac{1}{2\sqrt{6}} x^2 e^{-x/2}. \quad (10.96)$$

The spectrum of the hydrogen atom that we have found is shown in Fig. 10.6. The notation for the levels is $ns, np, \dots$ $1s$ denotes the ground state, $2s$ and $2p$ the first excited (degenerate) levels etc. All the levels are degenerate, except in the case $n = 1$. For a given value of n , all values of l lying between $l = 0$ and $l = n - 1$ are possible, and the degeneracy is

$$G(n) = \sum_{l=0}^{n-1} (2l+1) = n^2.$$

This degeneracy is peculiar to the Coulomb potential. The spectrum of the outer electron of an alkali atom (Fig. 10.7) qualitatively resembles that of the hydrogen atom, except that

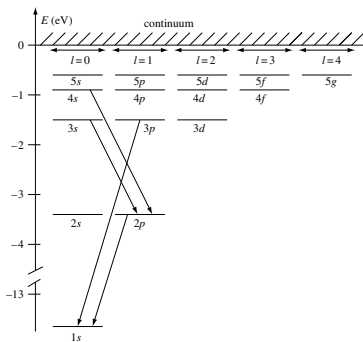


Fig. 10.6. Spectrum of the hydrogen atom.

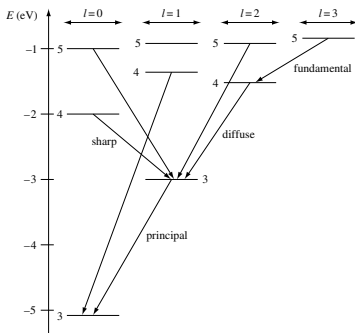


Fig. 10.7. Spectrum of the sodium atom.

there is no degeneracy. The Coulomb potential also presents a remarkable peculiarity in classical mechanics: it is the only potential, along with the harmonic potential $V(r) \propto r^2$, for which the trajectories close on themselves.⁹ This feature of the classical motion as well as the degeneracies associated with the quantum problem are due to the presence of an extra symmetry. This symmetry leads to an additional conservation law, that of the Lenz vector in the Coulomb case.

10.5 Angular distributions in decays

10.5.1 Rotations by π , parity, and reflection with respect to a plane

In this section we shall study decays of a particle C into two particles A and B :

$$C \rightarrow A + B. \quad (10.97)$$

We shall choose a reference frame in which particle C is at rest; particles A and B then have equal and opposite momenta $\vec{p}$ and $-\vec{p}$, respectively. The process (10.97) includes *radiative decays* (or *transitions*) with the emission of a photon, in which an excited level A^* of an atom, a molecule, or a nucleus emits a photon γ as the system undergoes a transition to a lower energy level A , which may or may not be the ground state:

$$A^* \rightarrow A + \gamma. \quad (10.98)$$

The states A^* and A may also correspond to different particles, as, for example, in the decay

$$\Sigma^0 \rightarrow \Lambda^0 + \gamma, \quad (10.99)$$

where the particles Σ^0 and Λ^0 are neutral particles formed from an up quark, a down quark, and a strange quark (Exercise 10.7.17).

The invariance under rotation of the Hamiltonian responsible for the decay implies conservation of angular momentum, which leads to constraints on the decay amplitudes and to important consequences for the angular distribution of the final particles. If the Hamiltonian governing the decay is invariant under parity, which is the case for the electromagnetic and strong interactions but not for weak interactions, we obtain additional constraints. It is convenient to introduce the operator $\mathcal{Y}$, the product of a rotation by π about the y axis and the parity operator Π (Section 8.3.3):

$$Y = e^{-i\pi J_y}, \quad \mathcal{Y} = Y \Pi = e^{-i\pi J_y} \Pi = \Pi e^{-i\pi J_y}. \quad (10.100)$$

This operator is just reflection with respect to the plane xOz ; $\mathcal{Y}$ is the reflection operator with respect to this plane. Let us first study the action of Y . This operator transforms J_x into $-J_x$ and J_z into $-J_z$ while leaving J_y unchanged:

$$Y^{-1} J_z Y = -J_z, \quad Y^{-1} J_{\pm} Y = -J_{\mp}. \quad (10.101)$$

⁹ The two cases are related; cf. Basdevant and Dalibard [2002], Chapter 11, Exercise 3. The extra symmetry can be used to find the energy levels and the wave functions, see e.g. E. Abers, *Quantum Mechanics*, New Jersey: Pearson Education (2004), Chapter 3.

Let us examine the action of Y on the state $|jm\rangle$:

$$J_z(Y|jm\rangle) = -YJ_z|jm\rangle = -m(Y|jm\rangle).$$

The state $Y|jm\rangle$ is then equal to $|j, -m\rangle$ up to a phase:

$$Y|jm\rangle = e^{i\alpha(j,m)}|j, -m\rangle,$$

because Y is unitary and preserves the norm. This result is not surprising, because the action of Y is equivalent to reversing the direction of the angular momentum quantization axis. Following the procedure used above in the case of parity, we apply J_+ to relate $\alpha(j, m)$ to $\alpha(j, m+1)$:

$$\begin{aligned} J_+Y|jm\rangle &= e^{i\alpha(j,m)}J_+|j, -m\rangle = \sqrt{j(j+1) - m(m-1)} e^{i\alpha(j,m)}|j, -m+1\rangle \\ &= -YJ_-|jm\rangle = -\sqrt{j(j+1) - m(m-1)} Y|j, m-1\rangle \\ &= -\sqrt{j(j+1) - m(m-1)} e^{i\alpha(j,m-1)}|j, -m+1\rangle, \end{aligned}$$

or

$$e^{i\alpha(j,m-1)} = -e^{i\alpha(j,m)}.$$

Since Y is a rotation by π , Y^2 is a rotation by 2π , $Y^2 = (-1)^{2j}$, and

$$Y^2|jm\rangle = e^{i\alpha(j,m)}e^{i\alpha(j,-m)}|jm\rangle = e^{2i\alpha(j,m)}(-1)^{2m}|jm\rangle = (-1)^{2j}|jm\rangle,$$

from which we find the two possible solutions

$$e^{i\alpha(j,m)} = (-1)^{j-m} \text{ or } e^{i\alpha(j,m)} = (-1)^{j+m}.$$

These two solutions are identical for integer j , while for $j = 1/2$ we can check using (10.38) that the first solution is the good one. It can be shown that this is also the case for all half-integer j . In the end, we have

$$\boxed{Y|jm\rangle = (-1)^{j-m}|j, -m\rangle, \quad Y^{-1}|jm\rangle = (-1)^{j+m}|j, -m\rangle}. \quad (10.102)$$

10.5.2 Dipole transitions

Now let us study radiative transitions of the type (10.98). First we return to the description of the photon polarization studied in Chapter 3, placing it within the general context of angular momentum. We have determined the infinitesimal generator of rotations of the polarization when the rotation is made about the propagation direction, taken to be the z axis. In the basis of linear polarization states $|x\rangle$ and $|y\rangle$ this infinitesimal generator is given by (3.26):

$$\Sigma_z = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

We have already seen in (3.29) that $\exp(-i\theta\Sigma_z)$ performs a rotation of the polarization in the xOy plane by an angle θ , and we can identify Σ_z as the z component of the photon angular momentum: $\Sigma_z = J_z$. Then according to (3.27) the action of the operator $\exp(-i\theta\Sigma_z)$ on the right- and left-handed polarization states $|R\rangle$ and $|L\rangle$ (3.11) is

$$\exp(-i\theta\Sigma_z)|R\rangle = e^{-i\theta}|R\rangle, \quad \exp(-i\theta\Sigma_z)|L\rangle = e^{i\theta}|L\rangle,$$

which proves that the states $|R\rangle$ and $|L\rangle$ have the magnetic quantum numbers $m = 1$ and $m = -1$, respectively.¹⁰ Furthermore, the description of the electromagnetic field by a vector potential shows that the photon has a vector nature and therefore spin 1, which permits $|R\rangle$ and $|L\rangle$ to be identified as the states $|jm\rangle$ (Fig. 10.8):

$$|R\rangle = |j = 1, m = 1\rangle = |11\rangle, \quad |L\rangle = |j = 1, m = -1\rangle = |1, -1\rangle, \quad (10.103)$$

where the angular momentum quantization axis Oz is taken to lie along the photon propagation direction. The value of m is called the photon *helicity*: $m = +1$ corresponds to positive helicity and $m = -1$ to negative helicity. Since angular momentum 1 corresponds to three possible values of the magnetic quantum number, $m = +1, 0, -1$, we might wonder what has happened to the value $m = 0$ for the photon. A general analysis due to Wigner shows that for a particle of zero mass and spin j , the only allowed eigenvalues of J_z are $m = j$ and $m = -j$, where the axis Oz is taken to lie along the particle propagation direction. When parity is not a symmetry of the Hamiltonian, the two possible values are independent. If the spin-1/2 neutrino had zero mass,¹¹ it would always have $m = -1/2$, while the antineutrino, which is a *different* particle, would always have $m = +1/2$. The photon interactions conserve parity as they are electromagnetic interactions, and so the same particle can have both $m = 1$ or $m = -1$.

We still need to check that the definition (10.103) corresponds to a standard angular momentum basis. We shall use the operator $Y = \exp(-i\pi J_y)$ which changes the direction

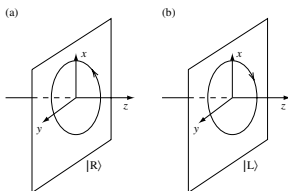
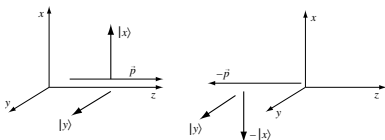


Fig. 10.8. (a) Right-handed circular polarization; (b) left-handed circular polarization.

¹⁰ An equivalent argument is to note that $\Sigma_z|R\rangle = |R\rangle$ and $\Sigma_z|L\rangle = -|L\rangle$.

¹¹ Which for a long time seemed possible, but apparently is not the case; see Exercise 4.3.6 and Footnote 4 of Chapter 1.

Fig. 10.9. Action of Y on linear polarization states.

of the photon propagation while leaving Oz unchanged. Its action on linear polarization states is (Fig. 10.9)

$$Y|x\rangle = -|x\rangle, \quad Y|y\rangle = |y\rangle.$$

We can derive its action on the circular polarization states $|R\rangle$ and $|L\rangle$ (3.11):

$$Y|R\rangle = Y\left[\frac{-1}{\sqrt{2}}(|x\rangle + i|y\rangle)\right] = \frac{1}{\sqrt{2}}(|x\rangle - i|y\rangle) = |L\rangle. \quad (10.104)$$

The relative phase of the states $|R\rangle$ and $|L\rangle$ corresponds to that of a standard basis since, according to (10.102),

$$Y|R\rangle = Y|1, 1\rangle = (-1)^{1-1}|1, -1\rangle = |L\rangle.$$

The choice (3.11) is also confirmed by the fact that $|R\rangle$ and $|L\rangle$ are given by the same combinations as the spherical components $\hat{r}_1$, $\hat{r}_{-1}$, and $\hat{r}_0$ (10.64) of $\hat{r}$.

Let us use $\vec{p}$ to denote the photon momentum, which we choose to lie along the z axis, and let $|jm\rangle$ be the angular momentum state of A^* (it is often said that the excited state has spin j), $|j'm'\rangle$ be the angular momentum state of the final level A (or the spin of the final level A), and $|\mu\rangle$ be the angular momentum state of the photon. Owing to the invariance under rotation, the angular momentum is conserved in the transition:

$$\vec{J} = \vec{J}' + \vec{S} + \vec{L},$$

where $\vec{S}$ is the photon spin and $\vec{L}$ is the orbital angular momentum. Projecting this equation on Oz , we find

$$m = m' + \mu + m_l.$$

It is easy to convince ourselves that the magnetic quantum number of the orbital angular momentum is zero: $m_l = 0$. In fact, the spatial wave function of the photon is a plane wave

$$e^{i\vec{p}\cdot\vec{r}} = e^{ipz} = e^{ipr\cos\theta},$$

which is invariant under rotation about Oz . The z component of the orbital angular momentum must be zero. Another justification follows from (10.47):

$$L_z e^{i\mu\phi} = -i \frac{\partial}{\partial \phi} e^{i\mu\phi} = 0.$$

The conservation of the angular momentum in the z direction gives

$$\begin{aligned} \text{right-handed final photon: } m &= m' + 1, \\ \text{left-handed final photon: } m &= m' - 1. \end{aligned} \quad (10.105)$$

If A and A^* have zero spin ($j = j' = 0$), then $m = m' = 0$ and the equations (10.105) have no solution: there is no single-photon radiative transition $j = 0 \rightarrow j' = 0$, often called a $0 \rightarrow 0$ transition. Radiative $0 \rightarrow 0$ transitions are possible only with the emission of at least two photons, and the probability of such a transition is suppressed by a power of the fine-structure constant $\alpha \simeq 1/137$.

A more interesting case which is often encountered in practice is that of $j = 1$ and $j' = 0$. If the photon is emitted in the z direction with helicity $\mu = \pm 1$, there are two possible cases taking into account $j' = m' = 0$:

$$\text{right-handed final photon: } m = 1, \quad \mu = 1, \quad (10.106)$$

$$\text{left-handed final photon: } m = -1, \quad \mu = -1. \quad (10.107)$$

Let a be the probability amplitude of (10.106) and b that of (10.107). It should be clearly understood that we are dealing with the amplitude of a *transition* probability, analogous to that calculated in (9.167), and not with probability amplitudes like those defined in postulate II of Chapter 4. The squared modulus of a transition amplitude gives the transition probability per unit time. The amplitudes a and b can be viewed as matrix elements of an operator T called the *transition matrix*, which can be calculated, at least formally, as a function of the Hamiltonian and which has the same symmetries as the Hamiltonian. We define the angle θ between the photon emission direction, taken to lie in the xOz plane, and the z axis, and we write the transition amplitudes a and b as (in (10.105) $m' = 0$ because $j' = 0$) for $\theta = 0$

$$\begin{aligned} a &= \langle R, \theta = 0 | T | j = 1, m = 1 \rangle = \langle R, \theta = 0 | T | 11 \rangle, \\ b &= \langle L, \theta = 0 | T | j = 1, m = -1 \rangle = \langle L, \theta = 0 | T | 1, -1 \rangle. \end{aligned} \quad (10.108)$$

If parity is a symmetry of the Hamiltonian responsible for the transition, then T commutes with $\mathcal{Y}$ (10.100). Since the two amplitudes a and b correspond to transitions which are deduced from each other by reflection with respect to the plane xOz (Fig. 10.10(a) and (b)), we must have $|a| = |b|$. To determine the phase in this relation we use

$$\begin{aligned} a &= \langle R, \theta = 0 | \mathcal{Y}^{-1} T \mathcal{Y} | 1, 1 \rangle = \eta_{\gamma} \eta_A \eta_{A^*} \langle L, \theta = 0 | T | 1, -1 \rangle \\ &= \eta_{\gamma} \eta_A \eta_{A^*} b, \end{aligned} \quad (10.109)$$

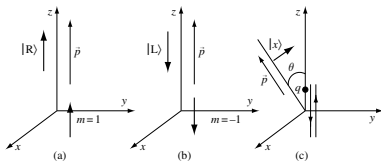


Fig. 10.10. Emission of photons with $\vec{p} \parallel Oz$. The amplitudes in (a) and (b) are deduced from each other by reflection with respect to the plane xOz . (c) Linear polarization of the final photon. The charge q undergoes oscillations along Oz .

where $\eta_X = \pm 1$ is the parity of the particle X . If X has momentum $\vec{p}$ and we write its state vector as $|X, \vec{p}\rangle$, then

$$\Pi|X, \vec{p}\rangle = \eta_X|X, -\vec{p}\rangle. \quad (10.110)$$

The description of the electromagnetic field by a vector potential, which is a polar vector, shows that the photon parity is $\eta_\gamma = -1$. Let $\eta = \eta_A \eta_{A^*}$. Then there are two possible cases:

1. $\eta = -1$, $a = b$;
2. $\eta = +1$, $a = -b$.

We are going to show that the first case is that of an electric dipole transition and the second is that of a magnetic dipole transition.¹² We do this by comparing with the simplest classical case, that of the radiation of a charge undergoing harmonic motion along the z axis. The classical angular momentum of this charge relative to the origin, and in particular its component in the z direction, is always zero, and the quantum case most similar to this situation is that where the excited state A^* possesses zero angular momentum in the z direction, that is, it is in the state $|j = 1, m = 0\rangle$. In order to compare the photon angular distribution with that of the classical radiation, we must imagine the case where the photon emission angle $\theta \neq 0$, the initial state of the atom being $|10\rangle$. We obtain the state $|R, \theta\rangle$ ($|L, \theta\rangle$) of the photon by rotation by an angle θ about Oy starting from $|R, \theta = 0\rangle$ ($|L, \theta = 0\rangle$):

$$|R, \theta\rangle = U[\mathcal{R}_y(\theta)]|R, \theta = 0\rangle,$$

$$|L, \theta\rangle = U[\mathcal{R}_y(\theta)]|L, \theta = 0\rangle.$$

¹² This result depends on the sign conventions used for the states $|R\rangle$ and $|L\rangle$; we find the sign opposite to that of Feynman *et al.* [1965], Vol III, Section 18.1 owing to the different sign convention in the definition of $|R\rangle$.

The emission amplitude in the θ direction, for example, for a right-handed photon and initial state $|j = 1, m = 0\rangle$, is

$$\begin{aligned}
 a_{\mathbf{R}}^{m=0}(\theta) &= \langle \mathbf{R}, \theta | T | 10 \rangle = \langle \mathbf{R}, \theta = 0 | U^\dagger [\mathcal{R}_{\hat{y}}(\theta)] T | 10 \rangle \\
 &= \langle \mathbf{R}, \theta = 0 | T U^\dagger [\mathcal{R}_{\hat{y}}(\theta)] | 10 \rangle \\
 &= \langle \mathbf{R}, \theta = 0 | T | 11 \rangle \langle 11 | U^\dagger [\mathcal{R}_{\hat{y}}(\theta)] | 10 \rangle \\
 &= a d_{01}^{(1)}(\theta) = \frac{a}{\sqrt{2}} \sin \theta.
 \end{aligned} \tag{10.111}$$

We have used the rotational invariance of T , introduced a set of intermediate states $\sum_m |1m\rangle \langle 1m|$ in the $j = 1$ subspace, and obtained the rotation matrix element using (10.39). A similar calculation gives the following for the emission of a left-handed photon:

$$a_{\mathbf{L}}^{m=0}(\theta) = b d_{0-1}^{(1)}(\theta) = -\frac{b}{\sqrt{2}} \sin \theta. \tag{10.112}$$

If the final polarization is linear, we can decompose it on the states $|x\rangle$ polarized in the plane xOz and $|y\rangle$ polarized along Oy (Fig. 10.10(c)): ¹³

$$|x\rangle = \frac{1}{\sqrt{2}}(-|R\rangle + |L\rangle), \quad |y\rangle = \frac{i}{\sqrt{2}}(|R\rangle + |L\rangle), \tag{10.113}$$

and we find

$$\begin{aligned}
 a_x^{m=0}(\theta) &= \langle x, \theta | T | 10 \rangle = -\frac{1}{2}(a + b) \sin \theta, \\
 a_y^{m=0}(\theta) &= \langle y, \theta | T | 10 \rangle = \frac{i}{2}(a - b) \sin \theta.
 \end{aligned} \tag{10.114}$$

In the electric dipole case $a = b$ the photons are polarized along Ox , while in the magnetic dipole case they are polarized along Oy . This corresponds to the classical case. If, for example, we take a charge undergoing harmonic oscillations along Oz with zero z component of angular momentum, the radiation is polarized in the plane xOz . On the other hand, a magnetic dipole will produce radiation polarized along Oy . An electric dipole transition corresponds to $\eta = -1$, and therefore to *initial and final states with opposite parities*, while a magnetic dipole transition corresponds to initial state and final state with the same parity. In both cases the angular distribution is $\sin^2 \theta$.

10.5.3 Two-body decays: the general case

Let us return to the general case of two-body decay (10.97), using j_A , j_B , and j_C to denote the spins of the particles A , B , and C . We define the transition amplitude for the initial

¹³ The states $|x\rangle$ and $|y\rangle$ are defined with respect to the propagation direction $\vec{p}$; see Fig. 10.10 (c).

where $\eta = \eta_A \eta_B \eta_C$ is the product of the parities of the three particles. Parity conservation halves the number of independent amplitudes. The amplitudes defined in (10.118) are called *helicity amplitudes*. However, it should be noted that the angular momentum quantization axis of particle B is often taken to be aligned with its momentum $-\hat{p}$, which causes $m_B \rightarrow -m_B$. The magnetic quantum numbers m_A and $-m_B$ (with our definition) are the helicities of particles A and B .

10.6 Addition of two angular momenta

10.6.1 Addition of two spins 1/2

In Section 6.1.2 we constructed a four-dimensional space $\mathcal{H}_1 \otimes \mathcal{H}_2$ by taking the tensor product of the two-dimensional spaces of two spins 1/2, $\vec{S}_1$ and $\vec{S}_2$. A possible basis in this space is formed from the eigenvectors $|\varepsilon_1 \varepsilon_2\rangle$, $\varepsilon = \pm$, of S_{1z} and S_{2z} :

$$|++\rangle, |+-\rangle, |-+\rangle, \text{ and } |--\rangle. \quad (10.120)$$

The physical properties that are diagonal in this basis are $\vec{S}_1^2$, $\vec{S}_2^2$, S_{1z} , and S_{2z} :

$$\vec{S}_1^2 |\varepsilon_1 \varepsilon_2\rangle = \frac{3}{4} |\varepsilon_1 \varepsilon_2\rangle, \quad S_{1z} |\varepsilon_1 \varepsilon_2\rangle = \varepsilon_1 |\varepsilon_1 \varepsilon_2\rangle, \quad (10.121)$$

$$\vec{S}_2^2 |\varepsilon_1 \varepsilon_2\rangle = \frac{3}{4} |\varepsilon_1 \varepsilon_2\rangle, \quad S_{2z} |\varepsilon_1 \varepsilon_2\rangle = \varepsilon_2 |\varepsilon_1 \varepsilon_2\rangle. \quad (10.122)$$

This basis corresponds to the following choice of complete set of compatible operators: $\{\vec{S}_1^2, \vec{S}_2^2, S_{1z}, S_{2z}\}$. It is possible to construct another interesting basis using the total angular momentum $\vec{S}$ obtained by adding $\vec{S}_1$ and $\vec{S}_2$:

$$\vec{S} = \vec{S}_1 + \vec{S}_2. \quad (10.123)$$

Here $\vec{S}$ is actually the total angular momentum, because it can be used to construct the infinitesimal generator in the tensor product space $\mathcal{H}_1 \otimes \mathcal{H}_2$ of a rotation $\mathcal{R}_{\hat{n}}(\theta)$ by an angle θ about the $\hat{n}$ axis:

$$U[\mathcal{R}_{\hat{n}}(\theta)] = e^{-i\theta(\vec{S}_1 \cdot \hat{n})} e^{-i\theta(\vec{S}_2 \cdot \hat{n})} = e^{-i\theta(\vec{S} \cdot \hat{n})}, \quad (10.124)$$

where we have used $[\vec{S}_1, \vec{S}_2] = 0$. Since $\vec{S}_1^2$ and $\vec{S}_2^2$ are scalar operators, they commute with $\vec{S}$, and another set of compatible operators is $\{\vec{S}_1^2, \vec{S}_2^2, \vec{S}^2, S_z\}$. We shall show below that this set is also complete. Let us find the basis vectors of this new set. Setting $|1, 1\rangle = |++\rangle$, we can show that

$$S_z |1, 1\rangle = |1, 1\rangle,$$

$$S_+ |1, 1\rangle = (S_{1+} + S_{2+}) |++\rangle = 0,$$

$$S_- |1, 1\rangle = (S_{1-} + S_{2-}) |++\rangle = |+-\rangle + |-+\rangle = \sqrt{2} |1, 0\rangle.$$

This last equation defines the normalized state vector $|1, 0\rangle$, which satisfies

$$S_z|1, 0\rangle = (S_{1z} + S_{2z})\frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle) = 0.$$

Finally,

$$S_-|1, 0\rangle = (S_{1-} + S_{2-})\frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle) = \sqrt{2}|- -\rangle = \sqrt{2}|1, -1\rangle,$$

$$S_z|1, -1\rangle = -|1, -1\rangle, \quad S_-|1, -1\rangle = 0.$$

These equations show that the three state vectors

$$\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$$

form a standard basis for angular momentum 1. It is sufficient to check the properties of the standard basis for S_z and S_- , because $S_+ = S_-^\dagger$ and $\vec{S}^2 = \frac{1}{2}(S_+S_- + S_-S_+) + S_z^2$. The above calculation shows that we have indeed constructed a standard basis; for example,

$$S_-|1, 1\rangle = \sqrt{j(j+1) - m(m-1)}|1, 0\rangle = \sqrt{2}|1, 0\rangle.$$

Finally, to obtain a basis of $\mathcal{H}_1 \otimes \mathcal{H}_2$, we need to construct a fourth vector orthogonal to the other three:

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle).$$

This vector is just the vector $|\Phi\rangle$ (6.15). As it is invariant under rotation, it corresponds to angular momentum zero, and it can be verified explicitly that

$$S_z|0, 0\rangle = 0, \quad S_\pm|0, 0\rangle = 0.$$

In summary, when two angular momenta $1/2$ are added, we obtain the angular momenta $s = 1$ and $s = 0$. A standard basis of $\vec{S}^2$ and S_z is formed from the vectors corresponding to $s = 1$:

$$s = 1 \quad \begin{cases} |1, 1\rangle = |++\rangle, \\ |1, 0\rangle = \frac{1}{\sqrt{2}}(|+-\rangle + |-+\rangle), \\ |1, -1\rangle = |--\rangle, \end{cases} \quad (10.125)$$

and $s = 0$:

$$s = 0 \quad |0, 0\rangle = \frac{1}{\sqrt{2}}(|+-\rangle - |-+\rangle). \quad (10.126)$$

Since we have found four orthogonal vectors, they form a basis of $\mathcal{H}_1 \otimes \mathcal{H}_2$, and the set of compatible operators $\{\vec{S}_1^2, \vec{S}_2^2, \vec{S}^2, S_z\}$, or simply $\{\vec{S}^2, S_z\}$, is complete. The $s = 1$ states are called *triplet states* and the $s = 0$ state is called the *singlet state*.

As an application, let us rederive the results of Exercise 6.5.4, where we diagonalized the operator $(\vec{\sigma}_1 \cdot \vec{\sigma}_2)$. This operator is diagonal in the basis $\{\vec{S}^2, S_z\}$. We have

$$\vec{S}^2 = \frac{1}{4}(\vec{\sigma}_1 + \vec{\sigma}_2)^2 = \frac{3}{2} + \frac{1}{2}\vec{\sigma}_1 \cdot \vec{\sigma}_2, \quad (10.127)$$

whence

$$\vec{\sigma}_1 \cdot \vec{\sigma}_2 = 2\vec{S}^2 - 3I = [2s(s+1) - 3]I.$$

The operator $\vec{\sigma}_1 \cdot \vec{\sigma}_2$ is equal to I in the triplet state and $-3I$ in the singlet state. We can find the projectors $\mathcal{P}_1$ and $\mathcal{P}_0$ on the triplet and singlet states:

$$\mathcal{P}_0 + \mathcal{P}_1 = I, \quad \vec{\sigma}_1 \cdot \vec{\sigma}_2 = -3\mathcal{P}_0 + \mathcal{P}_1,$$

from which

$$\boxed{\mathcal{P}_0 = \frac{1}{4}(I - \vec{\sigma}_1 \cdot \vec{\sigma}_2), \quad \mathcal{P}_1 = \frac{1}{4}(3 + \vec{\sigma}_1 \cdot \vec{\sigma}_2)} \quad (10.128)$$

The operator $\vec{\sigma}_1 \cdot \vec{\sigma}_2$ is a scalar operator which commutes with $\vec{S}$, but not with the individual spin operators $\vec{S}_1$ and $\vec{S}_2$. It should also be noted that the triplet states are symmetric (that is, they do not change sign) under the interchange of spins 1 and 2, while the singlet state is antisymmetric under this interchange.

10.6.2 The general case: addition of two angular momenta $\vec{J}_1$ and $\vec{J}_2$

Now let us generalize the preceding discussion to the addition of two angular momenta $\vec{J}_1$ and $\vec{J}_2$. The reasoning used in (10.124) can be repeated to show that $\vec{J} = \vec{J}_1 + \vec{J}_2$ is the total angular momentum. As in the preceding subsection, we construct the $(2j_1 + 1) \times (2j_2 + 1)$ -dimensional tensor product space:

$$\mathcal{E} = \mathcal{E}(j_1) \otimes \mathcal{E}(j_2).$$

A possible basis of this space is constructed from the eigenvectors

$$|j_1 j_2 m_1 m_2\rangle = |j_1 m_1\rangle \otimes |j_2 m_2\rangle \quad (10.129)$$

common to $\vec{J}_1^2$, $\vec{J}_2^2$, J_{1z} , and J_{2z} :

$$\vec{J}_1^2 |j_1 j_2 m_1 m_2\rangle = j_1(j_1 + 1) |j_1 j_2 m_1 m_2\rangle,$$

$$\vec{J}_2^2 |j_1 j_2 m_1 m_2\rangle = j_2(j_2 + 1) |j_1 j_2 m_1 m_2\rangle,$$

$$J_{1z} |j_1 j_2 m_1 m_2\rangle = m_1 |j_1 j_2 m_1 m_2\rangle,$$

$$J_{2z} |j_1 j_2 m_1 m_2\rangle = m_2 |j_1 j_2 m_1 m_2\rangle.$$

This basis corresponds to the complete set of commuting operators $\{\vec{J}_1^2, \vec{J}_2^2, J_{1z}, J_{2z}\}$. We shall construct another basis of $\mathcal{E}$ in which the operators $\{\vec{J}_1^2, \vec{J}_2^2, \vec{J}^2, J_z\}$ are diagonal. We start with the two following observations.

- Any vector $|j_1 j_2 m_1 m_2\rangle$ is an eigenvector of J_z with eigenvalue $m = m_1 + m_2$.
- If a value of j is allowed, by applying J_+ and J_- we generate a series of $(2j + 1)$ vectors $|jm\rangle$. A priori, we could have several series of vectors of this type, and we use $N(j)$ to denote the number of such series for a given value of j .

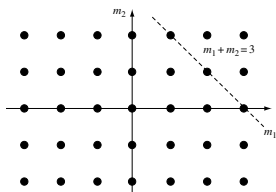


Fig. 10.12. Addition of two angular momenta.

Let $n(m)$ be the degeneracy of the eigenvalue m of J_z . Since m occurs if and only if $j \geq |m|$, we have (Fig. 10.12)

$$n(m) = \sum_{j \geq |m|} N(j),$$

and consequently

$$N(j) = n(j) - n(j+1).$$

However, $n(m)$ is equal to the number of pairs (m_1, m_2) such that $m = m_1 + m_2$. Assuming, for example, that $j_1 \geq j_2$,

$$n(m) = \begin{cases} 0 & \text{if } |m| > j_1 + j_2, \\ j_1 + j_2 + 1 - |m| & \text{if } j_1 - j_2 \leq m \leq j_1 + j_2, \\ 2j_2 + 1 & \text{if } 0 \leq |m| \leq j_1 - j_2. \end{cases}$$

We then conclude that

$$N(j) = 1 \quad \text{for } j_1 - j_2 \leq j \leq j_1 + j_2$$

and $N(j) = 0$ otherwise. To deal with the case $j_2 > j_1$ it is sufficient to replace $(j_1 - j_2)$ by $|j_1 - j_2|$. We can then state the following theorem.

The angular momentum addition theorem

In the tensor product space $\mathcal{E} = \mathcal{E}(j_1) \otimes \mathcal{E}(j_2)$

1. The possible values of j are

$$|j_1 - j_2|, |j_1 - j_2| + 1, \dots, j_1 + j_2 - 1, j_1 + j_2; \quad (10.130)$$

2. To each value of j there corresponds only one series of eigenvectors $|jm\rangle$:

$$\vec{J}^2 |jm\rangle = j(j+1) |jm\rangle, \quad J_z |jm\rangle = m |jm\rangle. \quad (10.131)$$

■ It is instructive to verify that the dimension of $\mathcal{E}$ is indeed correct ($j_1 \geq j_2$):

$$\begin{aligned}\dim \mathcal{E} &= \sum_{j_1 - j_2 \leq j \leq j_1 + j_2} (2j + 1) \\ &= (j_1 + j_2)(j_1 + j_2 + 1) - (j_1 - j_2 - 1)(j_1 - j_2) + (j_1 + j_2) - (j_1 - j_2 - 1) \\ &= (2j_1 + 1)(2j_2 + 1).\end{aligned}$$

Let us now go from the orthonormal basis $|j_1 j_2 m_1 m_2\rangle$ to the orthonormal basis $|jm\rangle$ by means of a unitary transformation. The elements of the unitary matrix that performs this transformation are called the *Clebsch–Gordan (CG) coefficients* $C_{m_1 m_2; jm}^{j_1 j_2}$:

$$|jm\rangle = \sum_{m_1 + m_2 = m} C_{m_1 m_2; jm}^{j_1 j_2} |j_1 j_2 m_1 m_2\rangle. \quad (10.132)$$

They can be nonzero only if $m = m_1 + m_2$ and $|j_1 - j_2| \leq j \leq j_1 + j_2$. We choose the following phase convention:

$$C_{m_1 m_2; j, m=j}^{j_1 j_2} \text{ real} \geq 0,$$

and then by application of J_- it can be shown that all the CG coefficients are real. The Clebsch–Gordan coefficients are the elements of a unitary real matrix with the matrix indices $(m_1 m_2)$ and (jm) . They therefore satisfy the orthogonality conditions

$$\sum_{m_1 = -j_1}^{j_1} \sum_{m_2 = -j_2}^{j_2} C_{m_1 m_2; jm}^{j_1 j_2} C_{m_1 m_2; j' m'}^{j_1 j_2} = \delta_{jj'} \delta_{mm'}, \quad (10.133)$$

and inversely

$$\sum_{j=|j_1-j_2|}^{j_1+j_2} \sum_{m=-j}^j C_{m_1 m_2; jm}^{j_1 j_2} C_{m'_1 m'_2; jm}^{j_1 j_2} = \delta_{m_1 m'_1} \delta_{m_2 m'_2}. \quad (10.134)$$

Equations (10.125) and (10.126) give examples of CG coefficients:

$$C_{\frac{1}{2} \frac{1}{2}; 11}^{\frac{1}{2} \frac{1}{2}} = 1, \quad C_{\frac{1}{2} -\frac{1}{2}; 10}^{\frac{1}{2} \frac{1}{2}} = \frac{1}{\sqrt{2}}.$$

As an application of angular momentum addition, let us study *spin–orbit coupling*. Owing to relativistic effects, the orbital angular momentum and the spin of an atomic electron, for example the electron of the hydrogen atom or the valence electron of an alkali atom, are not independent, as we shall see in Section 14.2.2. The total angular momentum of the electron is the sum of its orbital angular momentum $\vec{L}$ and its spin $\vec{S}$:

$$\vec{J} = \vec{L} + \vec{S}. \quad (10.135)$$

The possible values of j then are $j = l + 1/2$ and $j = l - 1/2$ (except if $l = 0$, in which case $j = s = 1/2$). The orbital angular momentum and the spin are coupled by a spin–orbit potential:

$$V_{so}(r) = V(r) \vec{L} \cdot \vec{S}. \quad (10.136)$$

This potential takes different values depending on whether $j = l + 1/2$ or $j = l - 1/2$. We can write

$$(\vec{L} + \vec{S})^2 = \vec{J}^2 = \vec{L}^2 + \vec{S}^2 + 2\vec{L} \cdot \vec{S}$$

and so

$$\vec{L} \cdot \vec{S} = \frac{1}{2} \left[j(j+1) - l(l+1) - s(s+1) \right], \quad (10.137)$$

which gives for the spin-orbit potential

$$V_{\text{so}}(r) = \begin{cases} \frac{1}{2} V(r) l & \text{for } j = l + 1/2, \\ -\frac{1}{2} V(r) (l+1) & \text{for } j = l - 1/2. \end{cases} \quad (10.138)$$

10.6.3 Composition of rotation matrices

The rule for the addition of angular momentum is reflected in a composition law for rotation matrices. Let us consider the matrix elements of the rotation operator $U(\mathcal{R})$ taken between states $|jm\rangle$ and $|jm'\rangle$ of the type (10.132):

$$\begin{aligned} \langle jm | U(\mathcal{R}) | jm' \rangle &= D_{mm'}^{(j)}(\mathcal{R}) \\ &= \sum_{m_1 m_2} \sum_{m'_1 m'_2} C_{m_1 m_2; jm}^{j_1 j_2} C_{m'_1 m'_2; jm'}^{j_1 j_2} \langle j_1 j_2 m_1 m_2 | U(\mathcal{R}) | j_1 j_2 m'_1 m'_2 \rangle, \end{aligned}$$

from which

$$D_{mm'}^{(j)}(\mathcal{R}) = \sum_{m_1 m_2} \sum_{m'_1 m'_2} C_{m_1 m_2; jm}^{j_1 j_2} C_{m'_1 m'_2; jm'}^{j_1 j_2} D_{m_1 m'_1}^{(j_1)}(\mathcal{R}) D_{m_2 m'_2}^{(j_2)}(\mathcal{R}). \quad (10.139)$$

Using the orthogonality relations (10.133) and (10.134) of the CG coefficients, we can invert (10.139):

$$D_{m_1 m'_1}^{(j_1)}(\mathcal{R}) D_{m_2 m'_2}^{(j_2)}(\mathcal{R}) = \sum_{|j_1 - j_2|}^{j_1 + j_2} C_{m_1 m_2; jm}^{j_1 j_2} C_{m'_1 m'_2; jm'}^{j_1 j_2} D_{mm'}^{(j)}(\mathcal{R}). \quad (10.140)$$

These equations can be interpreted in the following manner. In the space $\mathcal{E}(j_1) \otimes \mathcal{E}(j_2)$ we construct the matrix $\Delta(\mathcal{R})$, the tensor product of $D^{(j_1)}(\mathcal{R})$ and $D^{(j_2)}(\mathcal{R})$:

$$\Delta_{m_1 m_2; m'_1 m'_2}(\mathcal{R}) = D_{m_1 m'_1}^{(j_1)}(\mathcal{R}) \otimes D_{m_2 m'_2}^{(j_2)}(\mathcal{R}).$$

By a change of basis made using a unitary matrix whose elements are the CG coefficients $C_{m_1 m_2; jm}^{j_1 j_2}$, the matrix

$$\Delta'(\mathcal{R}) = C \Delta(\mathcal{R}) C^{-1}$$

becomes a block-diagonal matrix:

$$C\Delta(\mathcal{R})C^{-1} = \begin{pmatrix} D^{(j_1+j_2)} & 0 & \dots & 0 \\ 0 & D^{(j_1+j_2-1)} & \dots & \vdots \\ 0 & 0 & \dots & \vdots \\ 0 & \dots & 0 & D^{(|j_1-j_2|)} \end{pmatrix}.$$

In mathematical terms, this is referred to as reducing the product of two representations $D^{(j_1)}$ and $D^{(j_2)}$ of the rotation group to irreducible components:

$$D^{(j_1)} \otimes D^{(j_2)} = D^{(j_1+j_2)} \oplus D^{(j_1+j_2-1)} \oplus \dots \oplus D^{(|j_1-j_2|)}. \quad (10.141)$$

10.6.4 The Wigner-Eckart theorem (scalar and vector operators)

In Section 8.2.3 we defined a scalar operator $\mathcal{S}$ as an operator which commutes with $\vec{J}$: $[\mathcal{S}, \vec{J}] = 0$. Let us examine the matrix elements $\langle j' m' | \mathcal{S} | j m \rangle$ of $\mathcal{S}$ in a standard angular momentum basis:

$$[\mathcal{S}, \vec{J}^2] = 0 \Rightarrow j' = j, \quad [\mathcal{S}, J_z] = 0 \Rightarrow m' = m.$$

In addition,

$$[\mathcal{S}, J_{\pm}] = 0 \Rightarrow \langle j m | \mathcal{S} | j m \rangle = \langle j || \mathcal{S} || j \rangle \text{ is independent of } m. \quad (10.142)$$

The quantity $\langle j || \mathcal{S} || j \rangle$ is called the *reduced matrix element* of $\mathcal{S}$.

Now let us turn to vector operators $\vec{V}$, which we have defined in Section 8.2.3. The Cartesian components V_k of a vector operator transform under rotation as

$$U^\dagger(\mathcal{R}) V_k U(\mathcal{R}) = \sum_l \mathcal{R}_{kl} V_l. \quad (10.143)$$

By considering infinitesimal rotations, in Section 8.2.3 we derived the commutation relations involving the components of angular momentum:

$$[J_k, V_l] = i \sum_p \epsilon_{klp} V_p. \quad (10.144)$$

Equations (10.143) and (10.144) are strictly equivalent and either can be used to define a vector operator. It is convenient to use spherical components V_q of $\vec{V}$:

$$V_1 = -\frac{1}{\sqrt{2}}(V_x + iV_y), \quad V_0 = V_z, \quad V_{-1} = \frac{1}{\sqrt{2}}(V_x - iV_y). \quad (10.145)$$

These components are also called the standard components of $\vec{V}$, because when $\vec{V}$ is the position operator, $\vec{V} = \vec{R}$, the components $\hat{r}_1, \hat{r}_0$, and $\hat{r}_{-1}$ of the vector $\hat{r}$ are just the spherical harmonics $Y_1^\pm$ and Y_1^0 up to a factor of $\sqrt{3/4\pi}$ (cf. (10.64)). According to (10.65), this implies the transformation law

$$(\mathcal{R}\hat{r})_m = \sum_{m'} D_{m'm}^{(1)}(\mathcal{R}^{-1}) \hat{r}_{m'}. \quad (10.146)$$

The transformation law of the spherical components of $\vec{V}$ then is¹⁴

$$U(\mathcal{R})V_qU^\dagger(\mathcal{R}) = \sum_{q'} D_{q'q}^{(1)}(\mathcal{R})V_{q'}. \quad (10.147)$$

This can easily be checked using the explicit expressions for $D^{(1)}$ and the definition of the spherical components.

Our goal is to relate the matrix elements of the various components of a vector operator to the states $|jm\rangle$. To do this, let us study the properties of the vector $|1jqm\rangle = V_q|jm\rangle$ under rotation:

$$\begin{aligned} U(\mathcal{R})|1jqm\rangle &= U(\mathcal{R})V_qU^\dagger(\mathcal{R})U(\mathcal{R})|jm\rangle \\ &= \sum_{q'm'} D_{q'q}^{(1)}(\mathcal{R})D_{m'm}^{(j)}(\mathcal{R})|1jq'm'\rangle. \end{aligned}$$

The vectors $|1jqm\rangle$ transform under rotation in exactly the same way as the vectors $|j_1j_2m_1m_2\rangle$ with $j_1 = 1, j_2 = j, m_1 = q, m_2 = m$. We can then construct the vectors

$$|\bar{j}\bar{m}\rangle = \sum_{m+q=\bar{m}} C_{qm;\bar{j}\bar{m}}^{1j} |1jqm\rangle, \quad (10.148)$$

which transform under rotation as

$$U(\mathcal{R})|\bar{j}\bar{m}\rangle = \sum_{\bar{m}'} D_{\bar{m}'\bar{m}}^{(\bar{j})}(\mathcal{R})|\bar{j}\bar{m}'\rangle.$$

This equation shows that the vectors $|\bar{j}\bar{m}\rangle$ form a standard basis of the space $\mathcal{E}(\bar{j})$ up to a global multiplicative factor. These vectors will not in general be normalized, but they will have the same norm for any $\bar{m}$:

$$\langle \bar{j}\bar{m} | \bar{j}'\bar{m}' \rangle = \delta_{\bar{j}\bar{j}'} \delta_{\bar{m}\bar{m}'} \alpha(\bar{j}).$$

Inverting (10.148),

$$V_q|jm\rangle = |1jqm\rangle = \sum_{\bar{j}=j-1}^{j+1} C_{qm;\bar{j}\bar{m}}^{1j} |\bar{j}\bar{m}\rangle,$$

from which

$$\begin{aligned} \langle j'm' | V_q | jm \rangle &= \sum_{\bar{j}} C_{qm;\bar{j}\bar{m}}^{1j} \langle j'm' | 1j\bar{j}\bar{m} \rangle \\ &= \sum_{\bar{j}} C_{qm;\bar{j}\bar{m}}^{1j} \delta_{j'\bar{j}} \delta_{m'\bar{m}} \beta(j', j) = C_{qm;j',m'}^{1j} \beta(j', j). \end{aligned}$$

Defining the reduced matrix element $\langle j' || V_q || j \rangle$ as

$$\langle j' || V || j \rangle = \beta(j', j),$$

¹⁴ We note that the ordering of U and $U^\dagger$, as well as that of the indices, is different from that in (10.143).

we obtain the *Wigner–Eckart theorem* for vector operators:

$$\langle j' m' | V_q | j m \rangle = C_{qm; j' m'}^{1j} \langle j' || V || j \rangle. \quad (10.149)$$

All the dependence on the magnetic quantum numbers m , m' , and q is contained in the Clebsch–Gordan coefficient $C_{qm; j' m'}^{1j}$, which can be looked up in tables. For fixed j , the only possible values of j' are $j' = j - 1$, j , $j + 1$. This theorem can be generalized to *irreducible tensor operators*; see Exercise 10.7.18.

As an application, let us calculate the matrix elements of a vector operator when $j = j'$, using the fact that $\vec{J}$ is a vector operator with matrix elements satisfying (10.149):

$$\langle j m' | J_q | j m \rangle = C_{qm; j m'}^{1j} \langle j || J || j \rangle.$$

This leads to a proportionality relation for the Cartesian components V_k :

$$\langle j m' | V_k | j m \rangle = K \langle j m' | J_k | j m \rangle.$$

To evaluate the constant K , we calculate the scalar product $\vec{J} \cdot \vec{V}$, which is a scalar operator:

$$\begin{aligned} \langle j m | (\vec{J} \cdot \vec{V}) | j m \rangle &= \sum_{k, m''} \langle j m | J_k | j m'' \rangle \langle j m'' | V_k | j m \rangle \\ &= K \sum_{k, m''} \langle j m | J_k | j m'' \rangle \langle j m'' | J_k | j m \rangle \\ &= K \langle j m | \vec{J}^2 | j m \rangle = K j(j+1). \end{aligned}$$

Combining these equations, we obtain for the matrix elements of V_k

$$\langle j m' | V_k | j m \rangle = \frac{1}{j(j+1)} \langle j || (\vec{J} \cdot \vec{V}) || j \rangle \langle j m' | J_k | j m \rangle. \quad (10.150)$$

Since $(\vec{J} \cdot \vec{V})$ is a scalar operator, $\langle j m | (\vec{J} \cdot \vec{V}) | j m \rangle$ is independent of m and equal to the reduced matrix element $\langle j || (\vec{J} \cdot \vec{V}) || j \rangle$.

10.7 Exercises

10.7.1 Properties of $\vec{J}$

Show by explicit calculation that $[\vec{J}^2, J_z] = 0$. Also verify the identities (10.5) to (10.9).

10.7.2 Rotation of angular momentum

Let $\mathcal{R}$ be a rotation (10.30) by angles (θ, ϕ) . Show that the vector

$$U(\mathcal{R}) | j m \rangle = e^{-i\phi J_z} e^{-i\theta J_y} | j m \rangle$$

is an eigenvector of the operator

$$J_x \sin \theta \cos \phi + J_y \sin \theta \sin \phi + J_z \cos \theta = \vec{J} \cdot \hat{n}$$

with eigenvalue m . Here $\hat{n}$ is the unit vector in the direction (θ, ϕ) . Hint: adapt (8.29).

10.7.3 Rotations (θ, ϕ)

Show that the rotation (10.30) $\mathcal{R}(\theta, \phi)$ can be written as

$$\mathcal{R}(\theta, \phi) = \mathcal{R}_{y'}(\theta) \mathcal{R}_z(\phi),$$

where Oy' is the axis obtained from Oy by a rotation by ϕ about Oz . Hint: show that

$$\mathcal{R}_{y'}(\theta) = \mathcal{R}_z(\phi) \mathcal{R}_y(\theta) \mathcal{R}_z(-\phi).$$

10.7.4 The angular momenta $j = \frac{1}{2}$ and $j = 1$

1. Use (10.23) to find the operators S_x , S_y , and S_z for spin $1/2$.
2. Again using (10.23), calculate the 3×3 matrix representations of J_x , J_y , and J_z for angular momentum $j = 1$.
3. Show that for $j = 1$, J_x , J_y , and J_z are related to the infinitesimal generators (8.26) T_x , T_y , and T_z by a unitary transformation which takes the Cartesian components of $\hat{r}$ to the spherical components (10.64): $J_i = U^\dagger T_i U$ with

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 0 & 1 \\ -i & 0 & -i \\ 0 & \sqrt{2} & 0 \end{pmatrix}.$$

4. Calculate the rotation matrix $d^{(1)}(\theta)$:

$$d^{(1)}(\theta) = \exp(-i\theta J_y),$$

and verify (10.39). Hint: show that $J_y^3 = J_y$.

10.7.5 Orbital angular momentum

1. Use the canonical commutation relations

$$[X_i, P_j] = i\delta_{ij}I$$

and the expression $\vec{L} = \vec{R} \times \vec{P}$ to show that

$$[L_x, L_y] = iL_z.$$

2. Prove Equations (10.47) to (10.49). Hint: show that for an infinitesimal rotation by an angle $d\alpha$ about Ox , the angles θ and ϕ vary by

$$d\theta = -\sin \phi d\alpha, \quad d\phi = -\frac{\cos \phi}{\tan \theta} d\alpha.$$

Find L_x and $L_y = i[L_x, L_z]$.

3. Since $L_z = -i\partial/\partial\phi$, the following Heisenberg inequality should be valid:

$$\Delta\phi\Delta L_z \geq \frac{1}{2}.$$

In an eigenstate of L_z where m is fixed $\Delta L_z = 0$, whereas $\Delta\phi \leq 2\pi$ since $0 \leq \phi \leq 2\pi$. The Heisenberg inequality is therefore violated in this state. Where is the flaw in this argument? Hint: see Exercise 7.4.3, question 2. Why does the argument of Exercise 9.7.1 break down?

10.7.6 Relation between the rotation matrices and the spherical harmonics

1. Let $\varphi(\vec{r}) = \varphi(x, y, z)$ be the wave function of a particle. Show that

$$(e^{-iaL_z}\varphi)(0, 0, z) = \varphi(0, 0, z)$$

and that if a particle is localized on the z axis, the z component of its orbital angular momentum is zero. Interpret this result qualitatively.

2. We assume that the orbital angular momentum of the particle is l and write the wave function as the product of a spherical harmonic and a radial wave function $g_l(r)$ depending only on $r = |\vec{r}|$:

$$\psi_{lm}(\vec{r}) = Y_l^m(\theta, \phi)g_l(r) = \langle \theta, \phi | lm \rangle g_l(r).$$

We are interested uniquely in the angular part. Using

$$|\theta, \phi\rangle = U(\mathcal{R})|\theta = 0, \phi = 0\rangle,$$

where $\mathcal{R}$ is a rotation by the angles (θ, ϕ) , show that

$$Y_l^m(\theta, \phi) \propto [D_{m0}^{(l)}(\theta, \phi)]^*.$$

It can be shown that the proportionality coefficient is $\sqrt{(2l+1)/(4\pi)}$:

$$Y_l^m(\theta, \phi) = \sqrt{\frac{2l+1}{4\pi}} [D_{m0}^{(l)}(\theta, \phi)]^*.$$

10.7.7 Independence of the energy from m

Assuming that the potential $V(r)$ is invariant under rotation, let ψ_{lm} be a solution of the time-independent Schrödinger equation:

$$H\psi_{lm} = E_{lm}\psi_{lm}.$$

Use the commutation relation $[L_+, H] = 0$ to show that the energy E_{lm} is in fact independent of m .

10.7.8 The spherical well

1. We are given a potential $V(\vec{r})$ which is spherically symmetric (see Fig. 12.4):

$$V(\vec{r}) = -V_0, \quad 0 \leq r \leq R, \\ = 0, \quad r > R,$$

called a spherical well. Find the equation giving the s -wave ($l=0$) bound states. Is there always a bound state? Compare with the case of a one-dimensional well.

2. The neutron-proton potential can be modeled by a spherical well of radius $R \simeq 2$ fm. There is a single neutron-proton bound state in the s -wave, namely, the deuteron,¹⁵ with binding energy $B \simeq 2.2$ MeV. Calculate the depth V_0 of the well needed for there to be just a single bound state. Compare V_0 with the binding energy and show that $V_0 \gg B$.
3. Find the s -wave energy levels of a particle in the potential

$$V(r) = \frac{A}{r^2} - \frac{B}{r}, \quad A, B > 0.$$

10.7.9 The hydrogen atom for $l \neq 0$

1. Write down the equation that generalizes (10.89) when the orbital angular momentum $l \neq 0$. Show that it is necessary to add to (10.91) the term

$$-l(l+1) \left[\frac{a_1}{x} + \sum_{k=1}^{\infty} a_{k+1} x^{k-1} \right].$$

2. Prove the recursion relation

$$a_{k+1} = \frac{2(\alpha k - 1)}{k(k+1) - l(l+1)} a_k$$

and derive

- $\alpha = \frac{1}{n}$,
- $k \geq l+1$,

so that $l+1 \leq k \leq n$. Show that the spectrum of the hydrogen atom is given by (10.93).

10.7.10 Matrix elements of a potential

The external electron of an atom is assumed to be in a p state ($l=1$). Its wave function is

$$\psi_{1m}(\vec{r}) = Y_1^m(\theta, \phi) \frac{u_1(r)}{r}.$$

It is placed in an external potential of the form

$$V(\vec{r}) = Ax^2 + By^2 - (A+B)z^2,$$

where A and B are constants.

¹⁵ In fact, the deuteron also has a small d -wave component.

1. Show *without calculation* that the matrix representing V in the basis $|lm\rangle$ has the form

$$V_{m'm} = \begin{pmatrix} \alpha & 0 & \beta \\ 0 & \gamma & 0 \\ \beta & 0 & \alpha \end{pmatrix},$$

where the rows and columns are arranged in the order m' , $m = 1, 0, -1$.

2. Determine the eigenvalues and eigenvectors of V . Show that $\langle L_z \rangle = 0$ in an eigenstate of V .
 3. Use (10.63) to calculate α , β , and γ explicitly as functions of A , B , and

$$I = \int_0^\infty |u_1(r)|^2 r^2 dr.$$

10.7.11 The radial equation in dimension $d = 2$

We wish to write the equivalent of (10.78) in two-dimensional space when the potential is rotationally invariant. The time-independent Schrödinger equation is

$$\left[-\frac{1}{2M} \nabla^2 + V(r) \right] \psi(\vec{r}) = E \psi(\vec{r}).$$

We use polar coordinates in the plane xOy :

$$x = r \cos \theta, \quad y = r \sin \theta.$$

We recall the expression for the Laplacian in polar coordinates:

$$\nabla^2 = \frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}$$

and the expression for the angular momentum

$$L_z = XP_y - YP_x = -i \frac{\partial}{\partial \theta}.$$

1. Show that the eigenfunctions of L_z have the form $\exp(im\theta)$.
 2. We seek solutions of the Schrödinger equation of the form

$$\psi_{nm}(\vec{r}) = \frac{1}{\sqrt{r}} e^{im\theta} u_{nm}(r).$$

Show that $u_{nm}(r)$ and E_{nm} satisfy the radial equation

$$\left[-\frac{1}{2M} \frac{d^2}{dr^2} + V(r) + \frac{m^2 + 1/4}{2Mr^2} \right] u_{nm}(r) = E u_{nm}(r).$$

What is the interpretation of n ? What is the behavior of $u_{nm}(r)$ when $r \rightarrow 0$?

10.7.12 Symmetry property of the matrices $d^{(j)}$

Using the operator Y (10.100), demonstrate the symmetry property of the rotation matrices $d^{(j)}(\beta)$:

$$d_{m'm}^{(j)}(\beta) = (-1)^{m-m'} d_{-m', -m}^{(j)}(-\beta).$$

10.7.13 Light scattering

1. Let us resume the study of the radiative transition $A^* \rightarrow A + \gamma$ with $j = j(A^*) = 1$ and $j' = j(A) = 0$. Show in the electric dipole case that the transition amplitudes are the following for an initial state $m = 1$ when circularly polarized photons are emitted in the plane xOz with momentum $\vec{p}$ making an angle θ with the z axis:

$$a_R^{m=1}(\theta) = \frac{1}{2}a(1 + \cos \theta),$$

$$a_L^{m=1}(\theta) = \frac{1}{2}a(1 - \cos \theta).$$

Generalize to the case where the photon is emitted in the direction (θ, ϕ) .

2. We assume that photons of momentum $\vec{p} \parallel Oz$ arrive on the atom in its ground state A . The atom absorbs a photon and makes a transition to its excited state A^* . It then returns to the ground state by emitting a photon in the plane xOz at an angle θ with respect to Oz . We use b to denote the absorption amplitude of a photon of right-handed circular polarization R :

$$b = \langle j = 1, m = 1 | T' | R \rangle.$$

Show that if the transitions are of the electric dipole kind, we also have

$$b = \langle j = 1, m = -1 | T' | L \rangle.$$

Let $c_{P \rightarrow P'}(\theta)$ be the transition amplitude for the scattering of the initial photon of circular polarization P ($P = R$ or L) at an angle θ with final polarization P' . Show that

$$c_{P \rightarrow P'}(\theta) = \frac{ab}{2}(1 \pm \cos \theta),$$

where the $+$ sign corresponds to $P = P'$ and the $-$ sign to $P \neq P'$. Derive the transition amplitudes for linear polarization $|x\rangle$ of the initial photon and linear polarization $|x'\rangle$ or $|y\rangle$ of the scattered photon, defined with respect to the photon propagation direction:

$$c_{x \rightarrow x'}(\theta) = ab \cos \theta,$$

$$c_{x \rightarrow y}(\theta) = 0.$$

Give a classical analogy which also leads to a $\cos^2 \theta$ angular distribution with radiation polarized in the plane xOz . Generalize to the case where the photon is emitted in the direction (θ, ϕ) .

10.7.14 Measurement of the Λ^0 magnetic moment

The Λ^0 is a particle of zero charge, mass $M \simeq 1115 \text{ MeV } c^{-2}$, spin $1/2$, and lifetime $\tau \simeq 2.5 \times 10^{-10} \text{ s}$. One of its principal decay modes (66% of cases) is

$$\Lambda^0 \rightarrow \text{proton} + \pi^- \text{ meson},$$

where the proton has spin $1/2$ and the π^- meson has spin 0 .

1. In the reference frame where the Λ^0 is at rest, we assume that the proton is emitted with momentum $\vec{p}$ in the direction Oz , chosen to be the angular momentum quantization axis. Let m be the projection of the Λ^0 spin on the z direction and m' be that of the proton. Why must we have $m = m'$? Let a and b be the probability amplitudes of the transitions

$$a: \Lambda^0 \left(m = \frac{1}{2} \right) \rightarrow \text{proton} \left(m' = \frac{1}{2}; \vec{p} \parallel Oz \right),$$

$$b: \Lambda^0 \left(m = -\frac{1}{2} \right) \rightarrow \text{proton} \left(m' = -\frac{1}{2}; \vec{p} \parallel Oz \right).$$

Show that $|a| = |b|$ if parity is conserved in the decay. Hint: examine the action of a reflection with respect to the plane xOz .

2. The proton is now emitted with momentum $\vec{p}$ in the plane xOz parallel to the direction $\hat{n}$ making an angle θ with Oz . Let m' be the projection of the proton spin on the direction $\hat{n}$ and $a_{m'm}(\theta)$ be the amplitude:

$$a_{m'm}(\theta): \Lambda^0 \left(m = \frac{1}{2} \right) \rightarrow \text{proton} \left(m'; \vec{p} \parallel \hat{n} \right).$$

Express

$$a_{\frac{1}{2}, \frac{1}{2}}(\theta) = a_{++}(\theta) \quad \text{and} \quad a_{-\frac{1}{2}, \frac{1}{2}}(\theta) = a_{-+}(\theta)$$

as functions of a , b , and θ .

3. We assume that the Λ^0 is produced in the spin state $m = 1/2$. Show that the proton angular distribution is of the form

$$w(\theta) = w_0(1 + \alpha \cos \theta).$$

Calculate α as a function of a and b . Experiment shows that

$$\alpha \simeq -0.645 \pm 0.016.$$

What can be concluded about parity conservation in the decay?

4. The Λ^0 is produced by bombarding a target of protons at rest by a π^- -meson beam in the reaction (Fig. 10.13)

$$\pi^- \text{ meson} + \text{proton} \rightarrow \Lambda^0 + K^0 \text{ meson}.$$

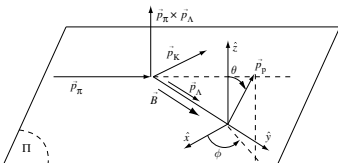


Fig. 10.13. Kinematics of Λ^0 production.

By momentum conservation, $\vec{p}_{\pi^-}$, $\vec{p}_{\Lambda^0}$, and $\vec{p}_{K^0}$ are located in the same plane. We choose the axis Oz to be perpendicular to this plane:

$$\hat{z} = \frac{\vec{p}_{\pi^-} \times \vec{p}_{\Lambda^0}}{|\vec{p}_{\pi^-} \times \vec{p}_{\Lambda^0}|},$$

and the axis Oy to be the direction $\vec{p}_{\Lambda^0}$ of the Λ^0 momentum. Given that parity is conserved in the production reaction and that the target protons are not polarized, show that if $\vec{S}$ is the Λ^0 spin operator, then the average values of the components S_x and S_y are zero: $\langle S_x \rangle = \langle S_y \rangle = 0$.

5. To simplify the situation, we assume that¹⁶ $\langle S_z \rangle = 1/2$ and that all the Λ^0 have the same lifetime τ and decay at the same point. The system is located in a uniform, constant magnetic field $\vec{B}$ parallel to Oy . The Λ^0 possesses a magnetic moment $\vec{\mu}$ related to its spin $\vec{S}$ by the gyromagnetic ratio γ : $\vec{\mu} = \gamma\vec{S}$. Qualitatively describe the motion of the Λ^0 spin. Determine its orientation at the instant the decay occurs as a function of τ , B , and γ . Show that the angular distribution of the proton emitted in the decay is

$$w(\theta, \phi) = w_0(1 + \alpha \cos \Theta)$$

with

$$\cos \Theta = \cos \lambda \cos \theta + \sin \lambda \sin \theta \cos \phi,$$

where the angles θ and ϕ are the polar and azimuthal angles of the proton momentum. What is the value of the angle λ ? Show that determination of $w(\theta, \phi)$ allows measurement of the gyromagnetic ratio γ . Neglect the curvature of the proton trajectory due to the magnetic field as well as the transformations of angles due to the motion of the Λ^0 .

10.7.15 Production and decay of the ρ^+ meson

1. The ρ^+ meson is a particle of spin 1 which decays into two π mesons, particles of spin 0:

$$\rho^+ \rightarrow \pi^+ + \pi^0.$$

We choose a reference frame in which the ρ^+ meson is at rest, and assume that its spin is quantized on the z axis and that it is initially in the spin state $|1m\rangle$, $m = -1, 0, 1$. Let

$$a_m(\theta, \phi) = \langle \theta, \phi | T | 1m \rangle$$

be the transition amplitude for the decay of a ρ^+ meson in the initial state $|1m\rangle$ with emission of a π^+ meson in the direction characterized by the polar and azimuthal angles (θ, ϕ) . Show that it is possible to write

$$a_m(\theta, \phi) = a \left[D_{m0}^{(1)}(\theta, \phi) \right]^*.$$

What is the physical significance of a ? Find the angular distribution $W_m(\theta, \phi)$ of the π^+ meson, that is, the π^+ emission probability in the direction (θ, ϕ) when the ρ^+ meson is initially in the state $|1m\rangle$. Show that $W_m(\theta, \phi)$ is independent of ϕ (why?) and give its explicit expression as a function of θ for the three values $m = -1, 0, 1$.

¹⁶ In fact, $|\langle S_z \rangle| < 1/2$ and we should use the state operator formalism for spin 1/2; see Section 6.2.2, where the Bloch vector $\vec{b}$ is identified with $2\langle \vec{S} \rangle$.

2. If the initial state of the ρ^+ meson is a linear combination of the states $|1m\rangle$,

$$|\lambda\rangle = \sum_{m=-1,0,1} c_m |1m\rangle, \quad \sum_{m=-1,0,1} |c_m|^2 = 1,$$

what will the angular distribution $W_\lambda(\theta, \phi)$ be?

3. In general, the ρ^+ is not produced in a pure state, but in a mixture described by a state operator ρ :

$$\rho = \sum_\lambda \mathbf{p}_\lambda |\lambda\rangle \langle \lambda|, \quad \mathbf{p}_\lambda \geq 0, \quad \sum_\lambda \mathbf{p}_\lambda = 1.$$

Show that the angular distribution is then

$$W(\theta, \phi) = \rho_{00} \cos^2 \theta + \frac{1}{2} \sin^2 \theta (\rho_{11} + \rho_{-1,-1}) \\ + \frac{1}{\sqrt{2}} \sin 2\theta \operatorname{Re} (\rho_{-10} e^{-i\phi} - \rho_{10} e^{i\phi}) - \sin^2 \theta \operatorname{Re} (\rho_{1,-1} e^{2i\phi}).$$

4. The ρ^+ meson is produced in the reaction π^+ meson ($\vec{p}_1$) + proton ($\vec{p} = 0$) $\rightarrow \rho^+$ meson ($\vec{p}_2$) + proton ($\vec{p}_3$), where $\vec{p}_i$ denotes the particle momentum. We choose the normal $\hat{n}$ to the reaction plane as the z axis:

$$\hat{n} = \frac{\vec{p}_1 \times \vec{p}_2}{|\vec{p}_1 \times \vec{p}_2|}.$$

The parity Π is conserved in this reaction and we assume that the target protons are not polarized. Show that the expectation value $\langle \vec{J} \rangle = \operatorname{Tr}(\rho \vec{J})$ of the ρ^+ spin points in the direction $\hat{n}$: $\langle \vec{J} \rangle = c \hat{n}$. Show that

$$\operatorname{Tr}(\rho J_x) = \operatorname{Tr}(\rho J_y) = 0.$$

Use the fact that the kinematics of the production reaction is invariant under the operation

$$\mathcal{Z} = \Pi e^{-i\pi J_z}, \quad [\rho, \mathcal{Z}] = 0,$$

to show

$$\rho_{mm'} = (-1)^{m-m'} \rho_{m'm},$$

so that ρ in fact depends only on four *real* parameters and has a checkerboard pattern

$$\begin{pmatrix} \rho_{11} & 0 & \rho_{1,-1} \\ 0 & \rho_{00} & 0 \\ \rho_{1,-1}^* & 0 & \rho_{-1,-1} \end{pmatrix}.$$

10.7.16 Interaction of two dipoles

The interaction Hamiltonian of two magnetic dipoles carried by particles of spin 1/2 is written as

$$H = \frac{K}{r^3} [3(\vec{\sigma}_1 \cdot \hat{r})(\vec{\sigma}_2 \cdot \hat{r}) - \vec{\sigma}_1 \cdot \vec{\sigma}_2] = \frac{K}{r^3} S_{12},$$

where $\vec{r}$ is the vector joining the two dipoles and $\vec{\sigma}_1$ and $\vec{\sigma}_2$ are the Pauli matrices of these particles. Let

$$\vec{\Sigma} = \frac{1}{2}(\vec{\sigma}_1 + \vec{\sigma}_2)$$

be the total spin. Show that

$$S_{12} = 2 \left(3Q^2 - \vec{\Sigma}^2 \right), \quad Q^2 = (\vec{\Sigma} \cdot \hat{r})^2,$$

and that $Q^4 = Q^2$, i.e., Q^2 is a projector. Show that $S_{12}^2 = 4\vec{\Sigma}^2 - 2S_{12}$ and that the eigenvalues of S_{12} are 0, 2, and -4 .

10.7.17 Σ^0 decay

The Σ^0 particle is composed of an up quark, a down quark, and a strange quark and has mass $1192 \text{ MeV } c^{-2}$ and spin $1/2$. It decays via a radiative transition to a Λ^0 particle, also composed of an up quark, a down quark, and a strange quark and having mass $1115 \text{ MeV } c^{-2}$ and spin $1/2$:

$$\Sigma^0 \rightarrow \Lambda^0 + \gamma.$$

The Σ^0 is assumed to be at rest, its spin is quantized along the z axis, and the spin projection on this axis is m . The photon momentum $\vec{p}$ lies in the plane xOz and makes an angle θ with the z axis.

1. First we assume that the photon is emitted in the z direction ($\theta = 0$). If m' is the projection of the Λ^0 spin on Oz , show that the nonzero amplitudes are (T is the transition operator)

$$a = \langle R, m' = -\frac{1}{2}; \theta = 0 | T | m = \frac{1}{2} \rangle,$$

$$b = \langle L, m' = \frac{1}{2}; \theta = 0 | T | m = -\frac{1}{2} \rangle,$$

while

$$c = \langle R, m' = \frac{1}{2}; \theta = 0 | T | m = \frac{1}{2} \rangle = 0,$$

$$d = \langle L, m' = -\frac{1}{2}; \theta = 0 | T | m = -\frac{1}{2} \rangle = 0,$$

in other words, $m' = m$ is forbidden and the allowed transitions correspond to $m' = -m$ when $\theta = 0$. The notation (R , L) specifies the right- or left-handed circular polarization state of the photon.

2. The transition operator T is invariant under the parity operation. Show that $|a| = |b|$. If η is the product of the Σ^0 and Λ^0 parities, also called the relative parity of the two particles

$$\eta = \eta_{\Sigma^0} \eta_{\Lambda^0},$$

show that $a = \eta b$. Experiment gives $\eta = 1$ and so $a = b$.

3. We assume that the initial value of the projection of the Σ^0 spin is $m = 1/2$. Let $a_R^{m'}(\theta)$ and $a_L^{m'}(\theta)$ be the transition amplitudes, where m' is the projection of the Λ^0 spin on the direction of $\vec{p}$, and therefore the eigenvalue of $\vec{S} \cdot \hat{p}$. Calculate $a_R^{m'}$ and $a_L^{m'}$ as functions of a and θ . What are the allowed values of m' ?

10.7.18 Irreducible tensor operators

An irreducible tensor operator of order k , $T^{(k)}$, possesses $(2k+1)$ components $T_q^{(k)}$:

$$q = -k, -k+1, \dots, k-1, k$$

and transforms under a rotation $\mathcal{R}$ as

$$U(\mathcal{R})T_q^{(k)}U^\dagger(\mathcal{R}) = \sum_{q'} D_{q'q}^{(k)}(\mathcal{R})T_{q'}^{(k)}.$$

Show that the vector

$$|kjqm\rangle = T_q^{(k)}|jm\rangle$$

transforms under rotation as the vector $|j_1j_2m_1m_2\rangle$ with $j_1 = k$, $j_2 = j$, $m_1 = q$, and $m_2 = m$. Using the vectors

$$|kj\tilde{j}\tilde{m}\rangle = \sum_{q+m=\tilde{m}} C_{qm;\tilde{j}\tilde{m}}^{kj} |kjqm\rangle$$

as intermediaries, prove the general form of the Wigner–Eckart theorem:

$$\langle j'm'|T_q^{(k)}|jm\rangle = C_{qm;j'm'}^{kj} \langle j'||T^{(k)}||jm\rangle$$

and show that

$$|j-k| \leq j' \leq j+k.$$

10.8 Further reading

The presentation in this chapter, inspired by that of Feynman *et al.* [1965], Vol. III, Chapters 17 and 18, places particular emphasis on the properties and use of the rotation matrices. For a more classical presentation the reader can consult Messiah [1999], Chapter XIII, Cohen-Tannoudji *et al.* [1977], Chapter VII, or Basdevant and Dalibard [2002], Chapter 10. Numerous applications to elementary particle physics can be found in the book by S. Gasiorowicz, *Elementary Particle Physics*, New York: Wiley (1966). In addition, Chapter 4 of that book describes the Wigner analysis based on invariance under the Poincaré group, which shows in particular that a particle of zero mass has only two helicity states, whatever its spin. On this last subject see also Weinberg [1995], Chapter 2.